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Preface

This tutorial aims to introduce two excellent Bitcoin Mining projects. Freenove has adapted and validated these projects to ensure stable operation on our Freenove ESP32 Display, and have accordingly compiled detailed usage tutorials.

Disclaimer:

Freenove is not the original developer of these projects. All development, maintenance, feature implementation, and commercial licensing are the sole responsibility of their respective official developers.

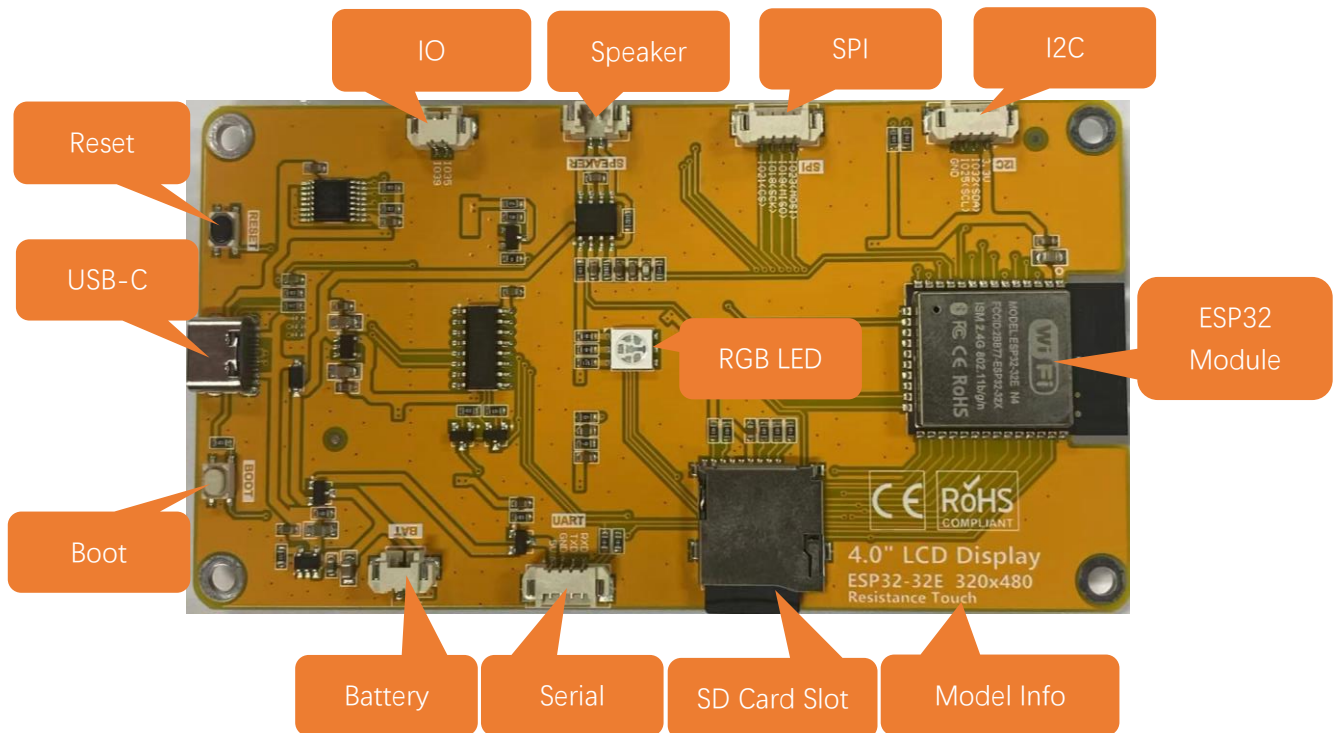
Miner Compatibility

Provide a tutorial to show how to install the following miner firmware.

Model	NMMiner (2.99 USD per device)	NerdMiner (free)
2.8 Inch ST7789 TN	 fully compatible	 fully compatible
2.8 Inch ILI9341 TN	 fully compatible	 fully compatible
3.2 Inch ST7789 IPS	 fully compatible, light theme	 slightly incompatible in display color
3.5 Inch ST7796 TN	 fully compatible	 incompatible in screen display
4.0 Inch ST7796 TN	 fully compatible	 incompatible in screen display

Freenove ESP32 Display

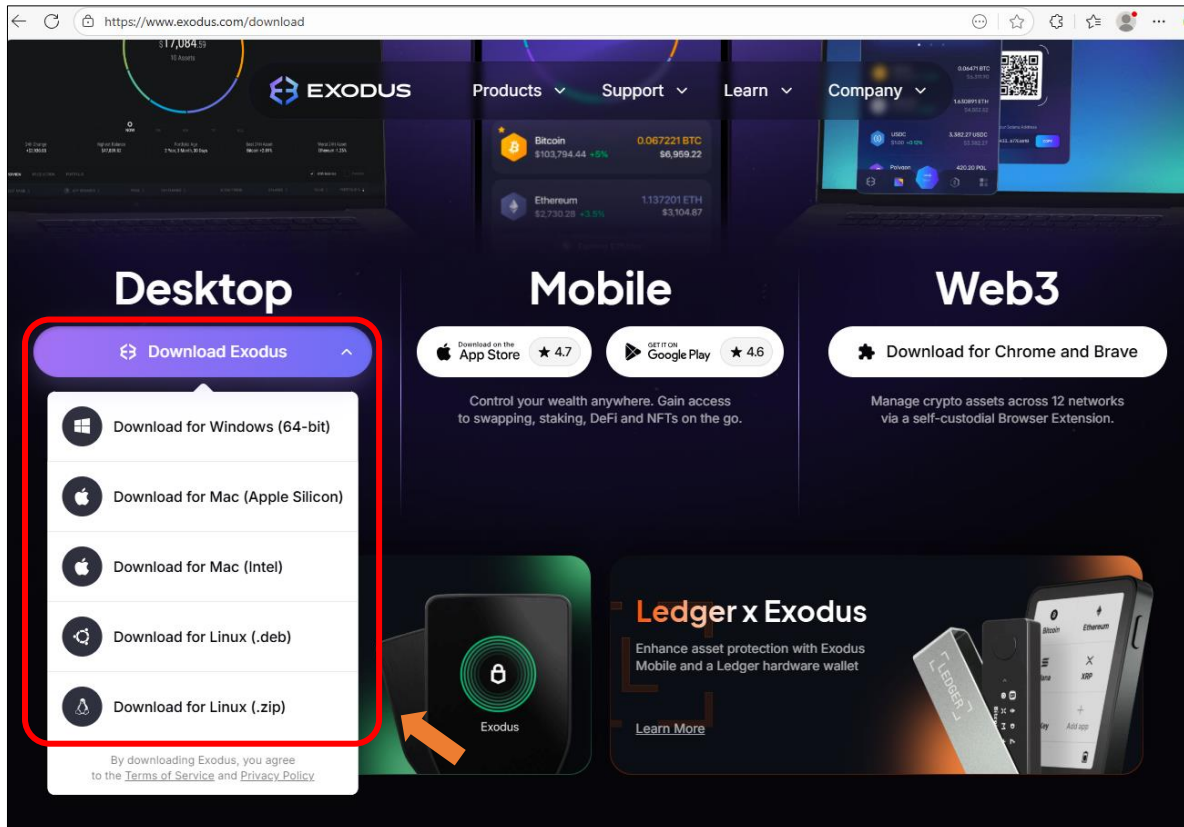
Hardware Interfaces



Obtaining the BTC Receiving Address

Note: If you already have a BTC receiving address, feel free to skip this step. Otherwise, the following section will guide you through obtaining a new one.

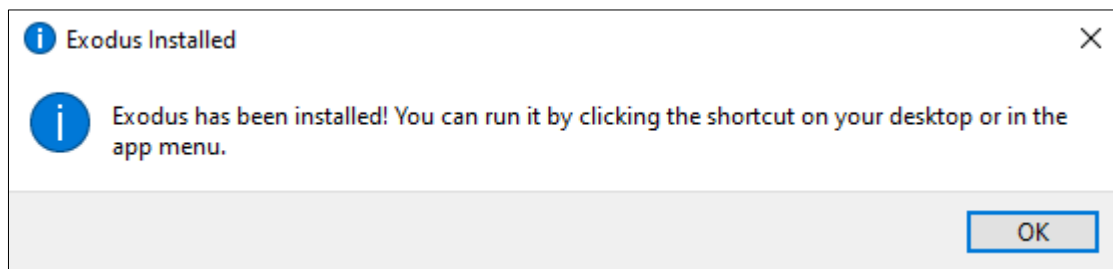
1. Click this link (<https://www.exodus.com/download>) to open the Exodus downloading interface. Choose the version consistent with your system.



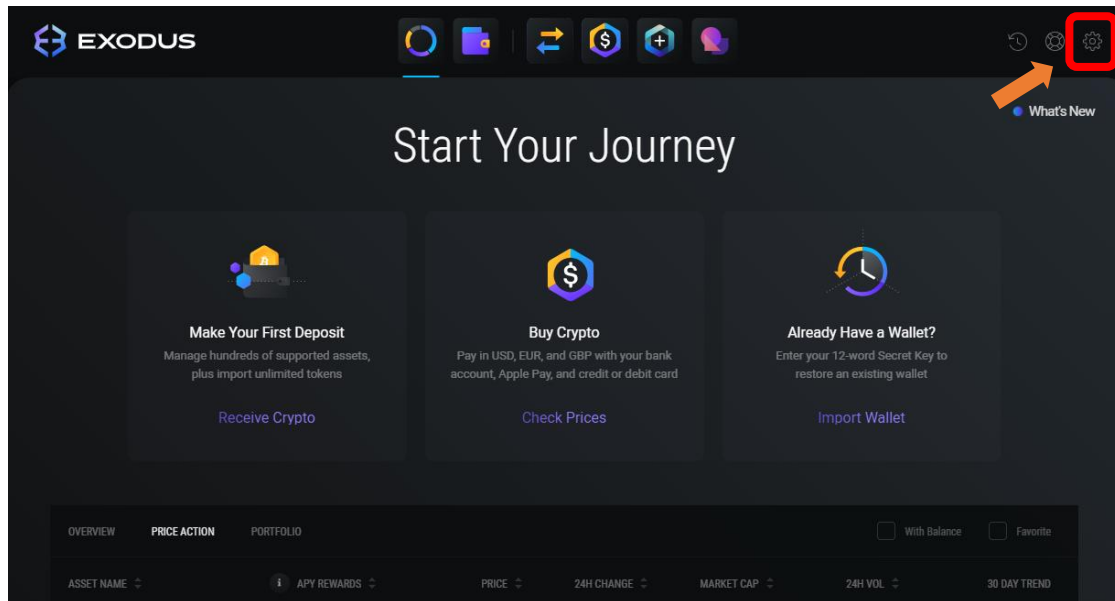
2. Open the downloaded installation package to install.



3. The following prompt indicates that the installation is successful.

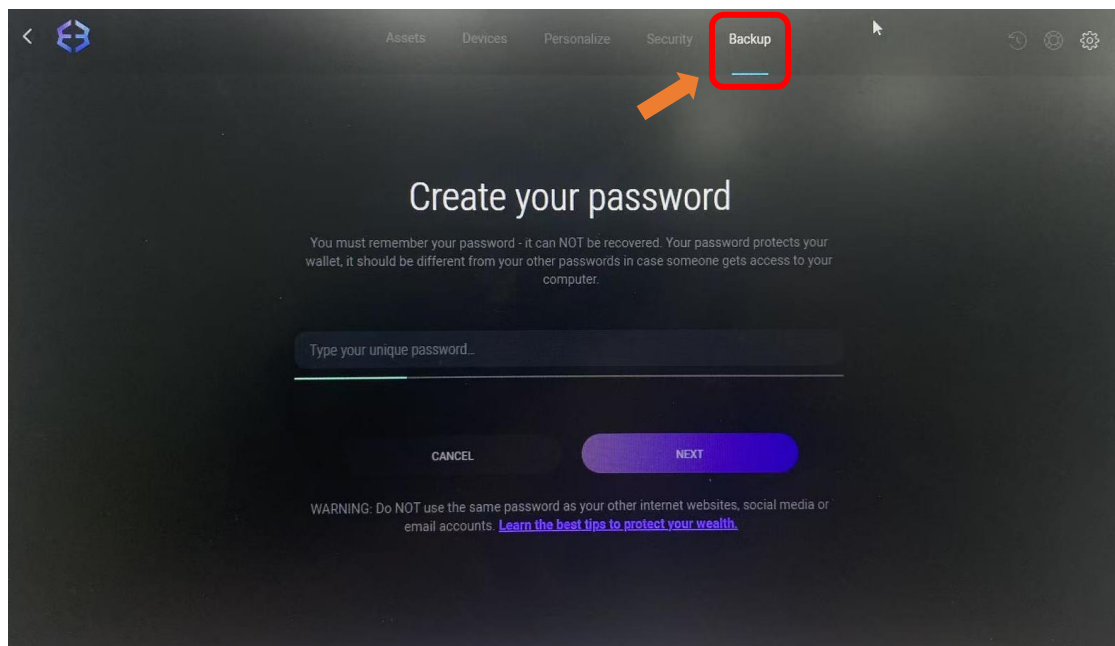


4. Open **Exodus**, click "**Setting**".

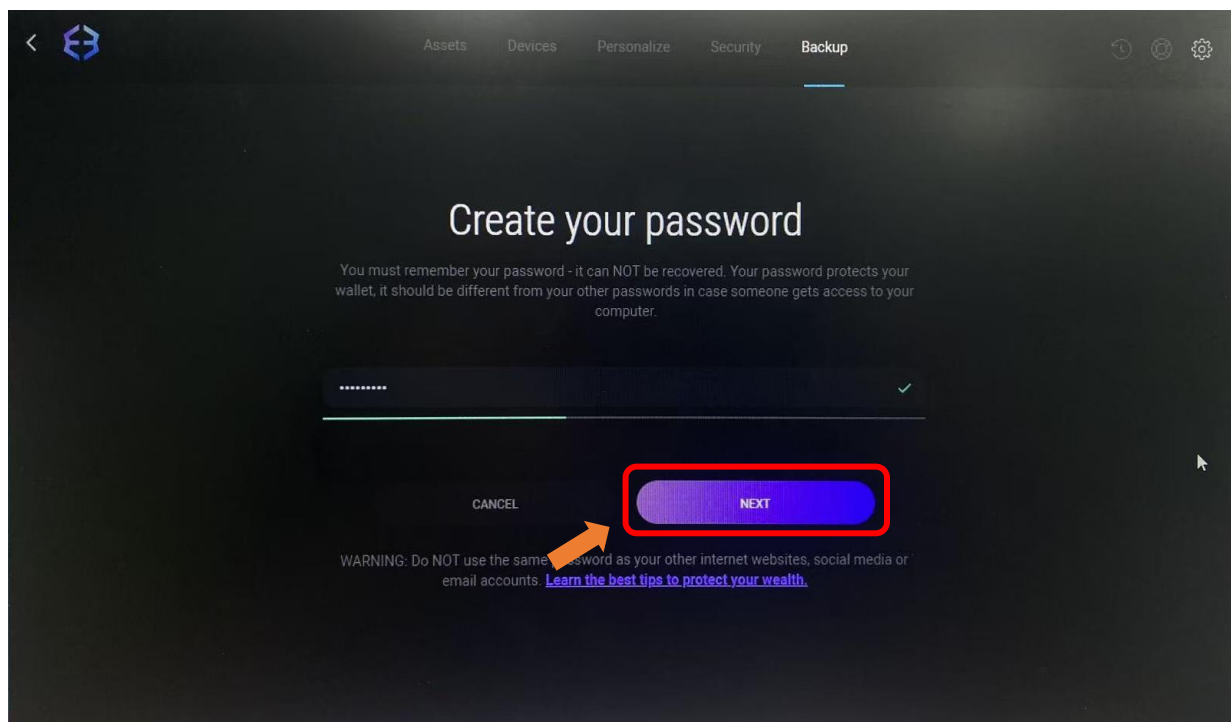


5. Click "**Backup**" to create your password for registering your wallet.

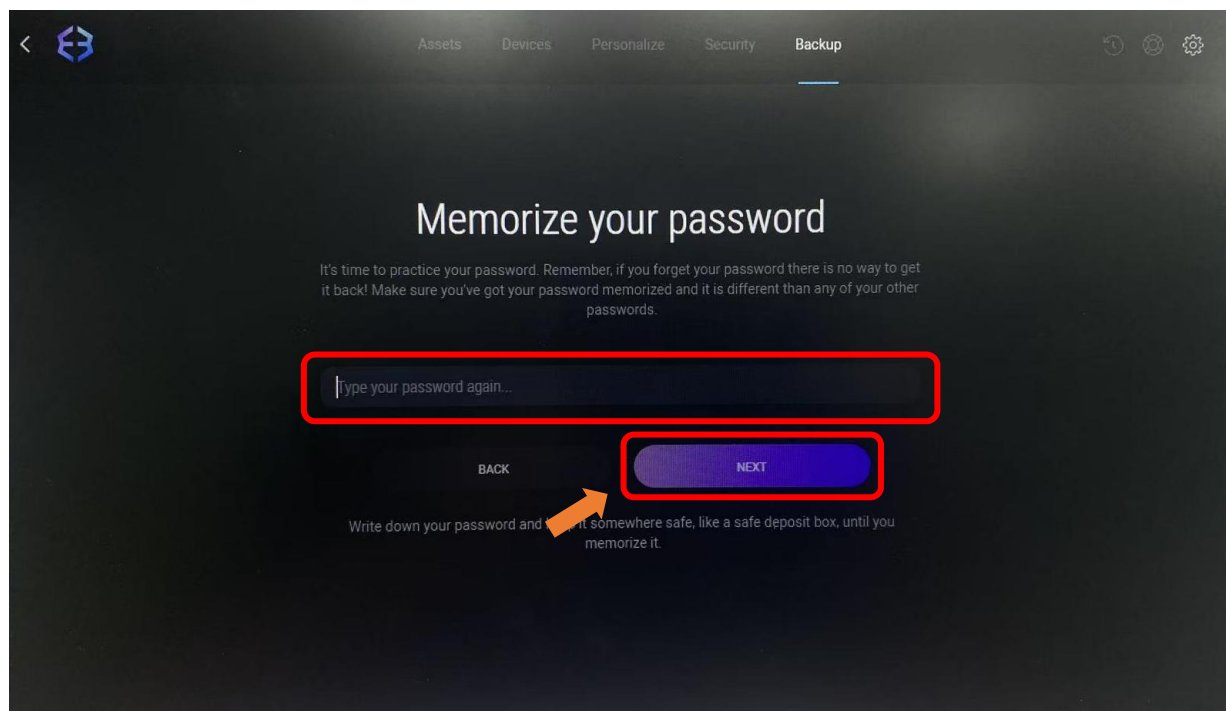
Important Note: Do not forget or share this password with anyone.



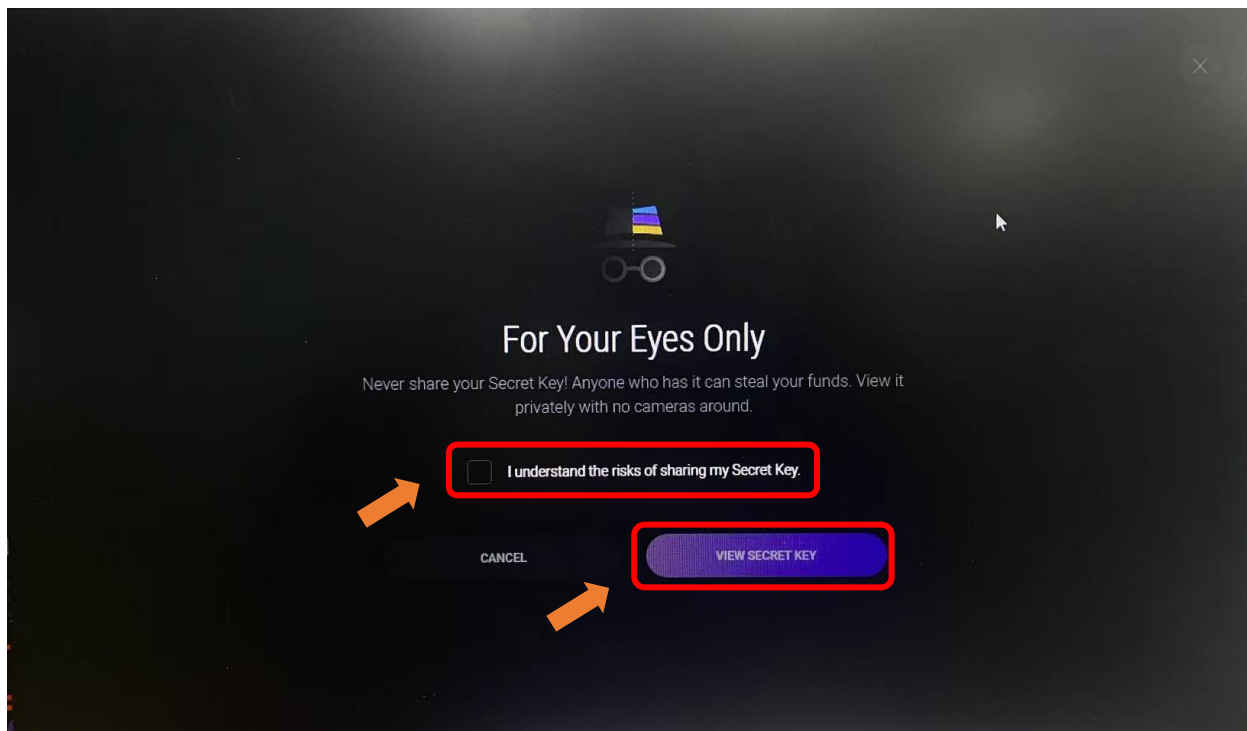
6. Click "NEXT".



7. Enter the password again and click "NEXT".



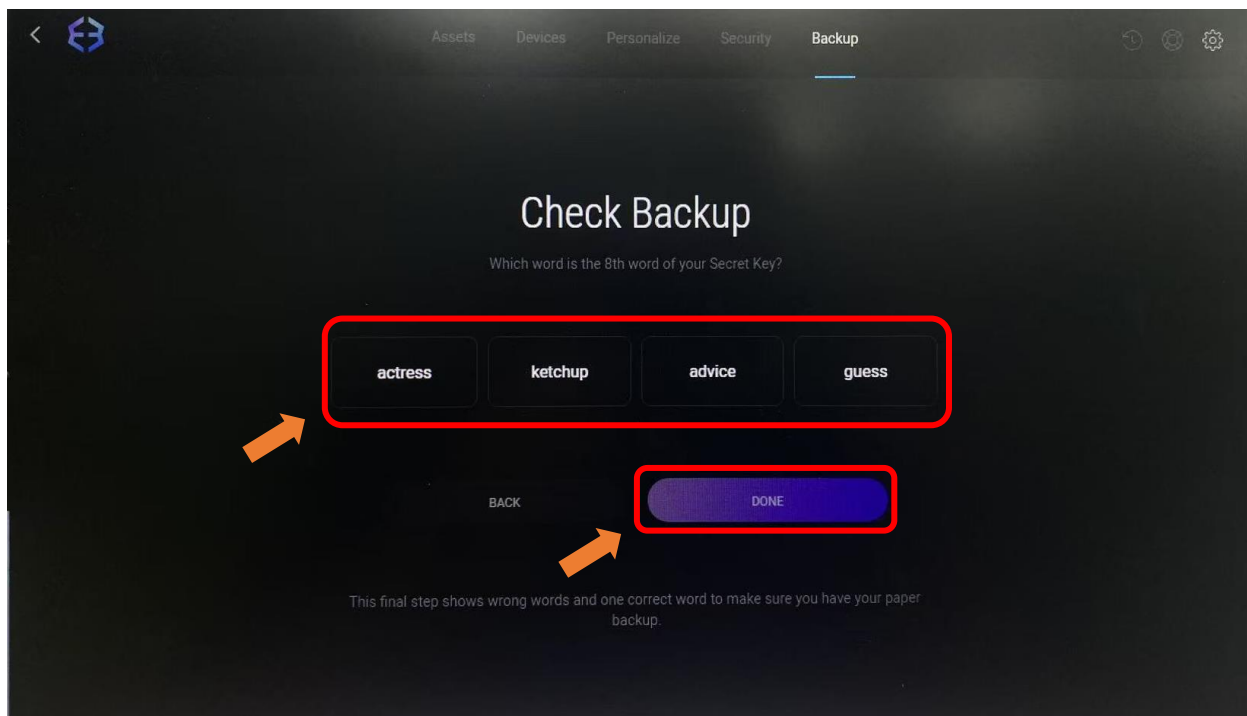
8. Check the box "I understand the risks of sharing my Secret Key", and click "VIEW SECRET KEY".



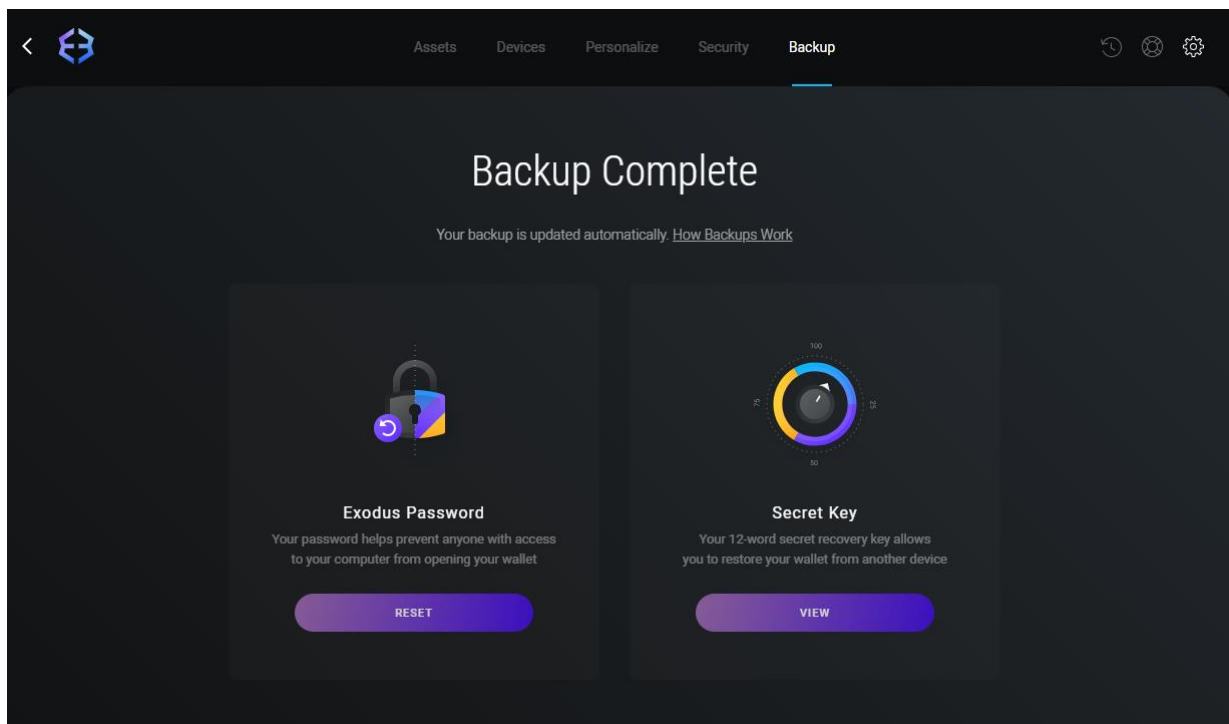
9. Check "Show Secret Key".

Please note: This interface cannot be screenshot or photographed. Do not disclose any information related to these mnemonic words to anyone. You must securely store these 12 mnemonic words corresponding to the numbers. Without them, you will permanently lose access to this wallet.

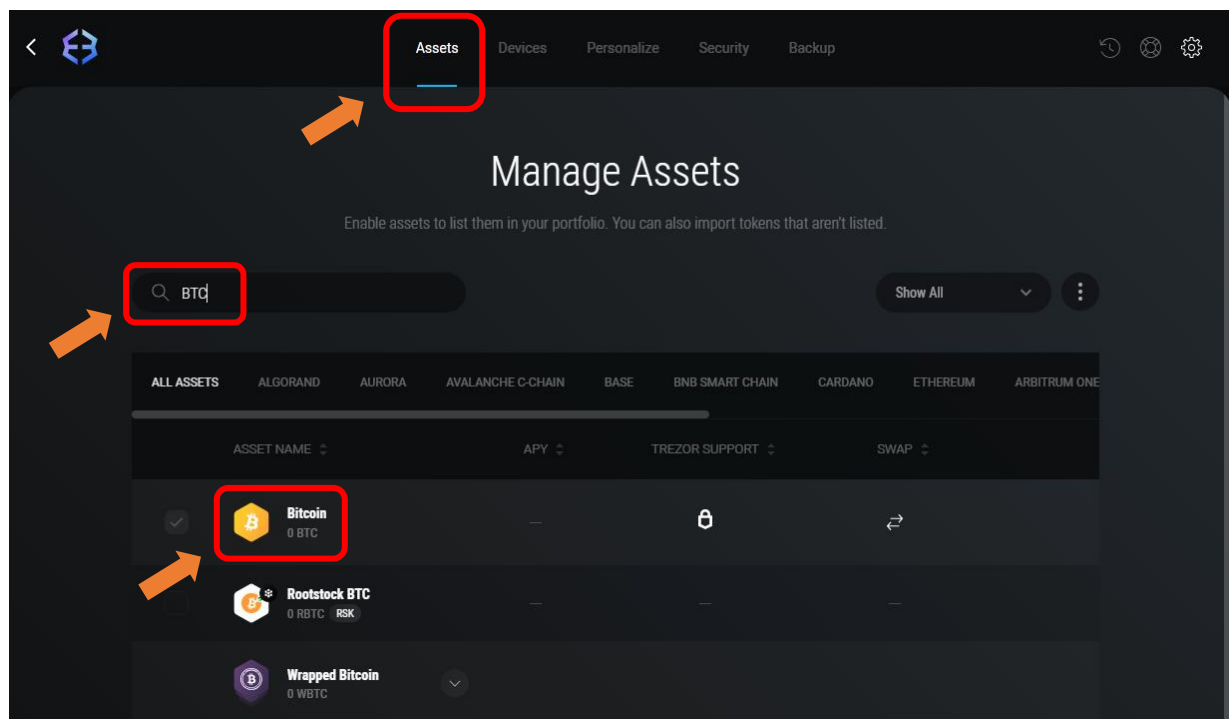
10. Click "DONE".
11. Select the correct answer based on the recorded mnemonic words. Then click "DONE".



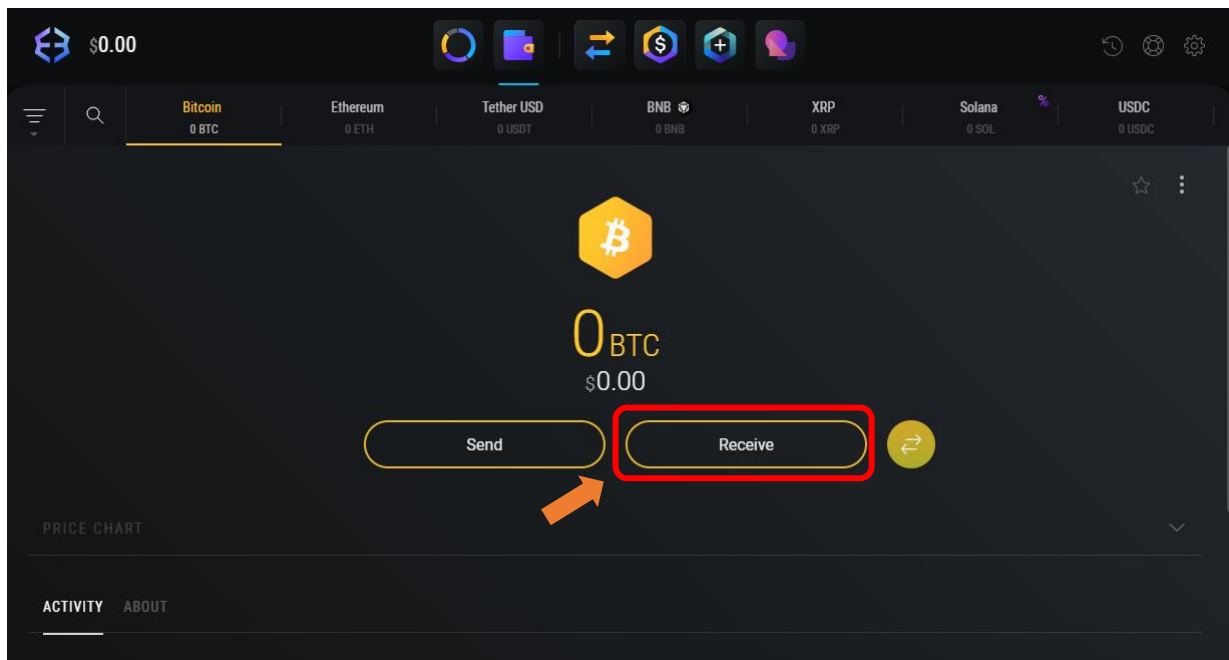
12. So far, you have successfully created your wallet.



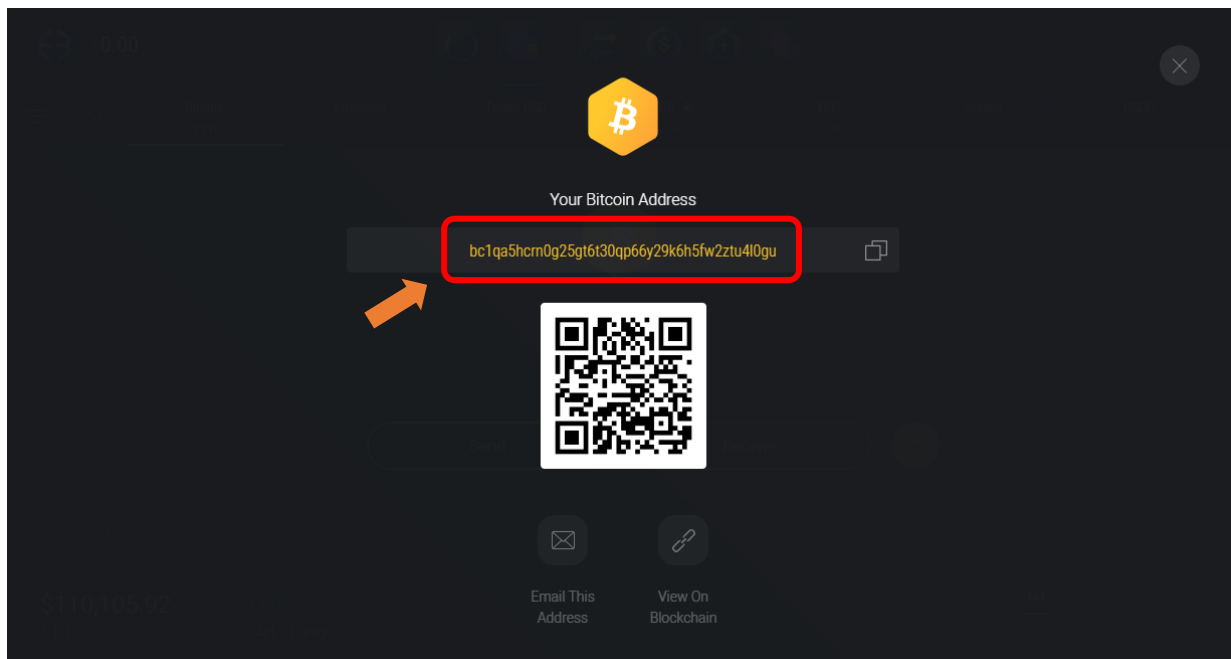
13. Click “Assets”, type “BTC” to search, and then click “Bitcoin”.



14. Click "Receive".



15. The generated string of characters is a BTC address. This address can be publicly shared and is used to receive BTC.



For any Exodus-related usage questions, answers can be found at:

<https://www.exodus.com/support/en/>

If you have any concerns, please feel free to contact us via support@freenove.com

NMMiner

This project utilizes Freenove ESP32 Display and NMMiner to implement Bitcoin mining. It requires some programming background and a basic understanding of digital currencies.

Project Introduction

NMMiner (<https://github.com/NMminer1024/NMMiner>) is an excellent BTC mining firmware specifically designed for ESP32 development boards. It breaks away from the traditional mining requirements of massive electricity consumption and high hash rates by adopting a SOLO mining mode. In this mode, every computation has the chance to independently win the entire block reward. NMMiner supports all cryptocurrencies based on the SHA256d algorithm, such as **BTC, BCH, XEC, and DGB**. Freenove has successfully ported this project to run on the Freenove ESP32 Display.

This tutorial will guide you through the process of running the project on the Freenove ESP32 Display.

NMMiner offers two management tools:

1. **NMController Client**: Windows desktop application
2. **NMController Web**: A cross-platform web client designed with Python, compatible with both Windows and macOS systems.

Precautions

- **Cost**
 - NMMiner is currently a paid service. Each device requires a unique license for activation. Therefore, you will need to purchase a separate activation code for every development board you plan to use in order to complete the activation process.
- **Security Precautions**
 - Never disclose your wallet's private key or recovery phrase to anyone. The only information you need to provide is your public cryptocurrency receiving address.
- **Troubleshooting**
 - If you have followed the tutorial but are still unable to use the miner, please feel free to contact us for assistance. (support@freenove.com)

Compatibility Description

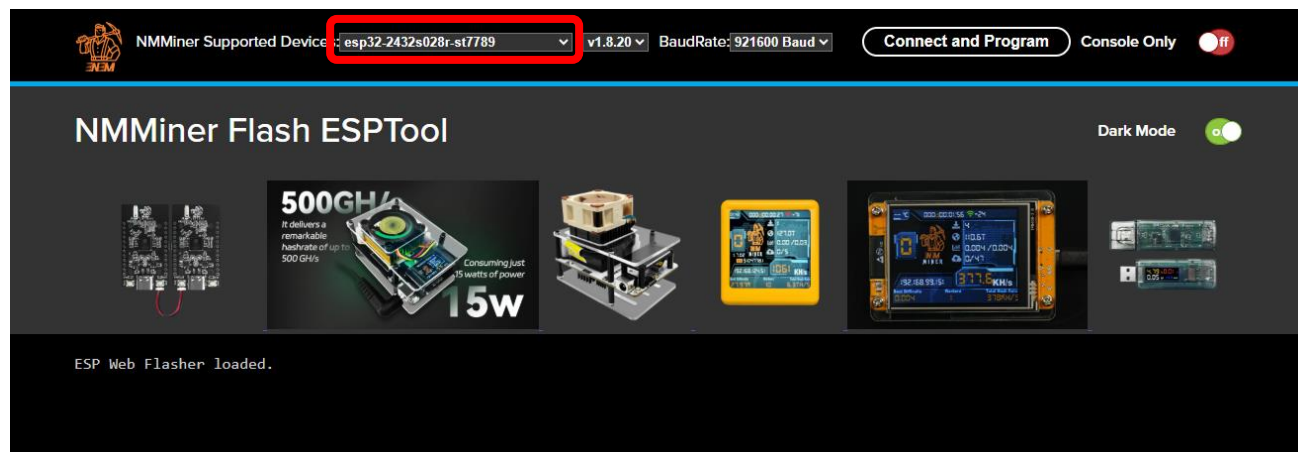
At the time of writing this guide, the Freenove ESP32 Display is available in five different models. Although they vary in terms of screen drivers, resolutions, and screen sizes, all of them can be set up using this tutorial to learn how to use NMMiner.

Models	Specifications		Effect
FNK0103F_2P8	Size	2.8 inch	Fully Compatible
	Resolution	240x320	
	Driver	ILI9341	
FNK0103B_2P8	Size	2.8inch	Fully Compatible
	Resolution	240x320	
	Driver	ST7789	
FNK0103L_3P2	Size	3.2inch	Color Inversion
	Resolution	240x320	
	Driver	ST7789	
FNK0103N_3P5	Size	3.5inch	Fully Compatible
	Resolution	320x480	
	Driver	ST7796	
FNK0103S_4P0	Size	4.0inch	Fully Compatible
	Resolution	320x480	
	Driver	ST7796	

How to Use

Firmware Flashing and Activating

1. Open the [NMMiner Flash ESPTool](#) interface, click to select the device.

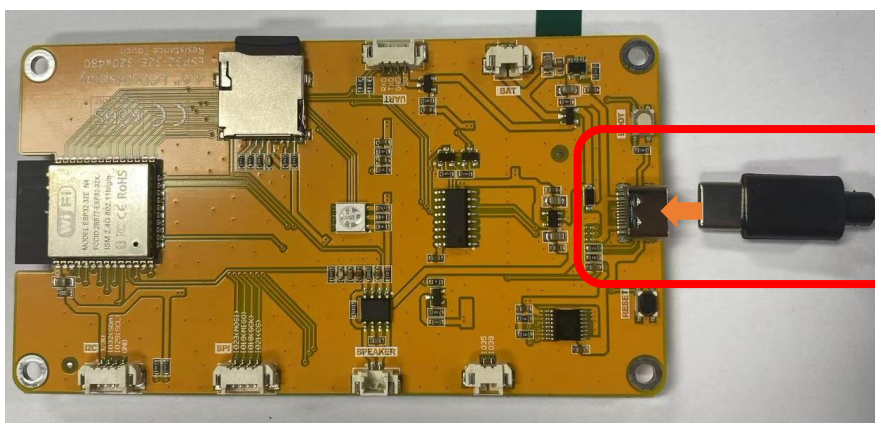


2. Select the device corresponding to your display according to the table below.

Models	Device Name
FNK0103F_2P8	esp32-2432s028r-ili9341
FNK0103B_2P8	esp32-2432s028r-st7789
FNK0103L_3P2	esp32-2432s024
FNK0103N_3P5	esp32-3248s035
FNK0103S_4P0	esp32-3248s035

If you have any concerns, please feel free to contact us via support@freenove.com

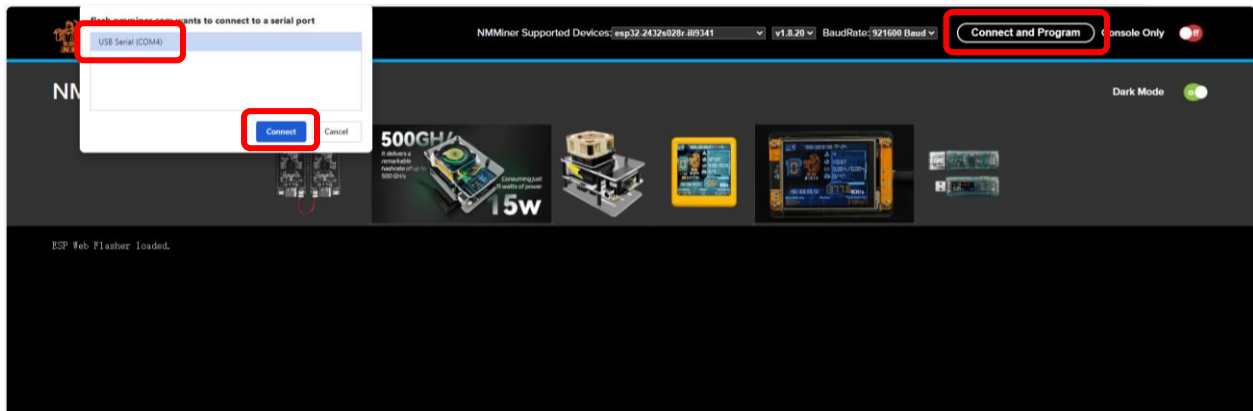
3. Connect the Freenove ESP32 Display to your computer.



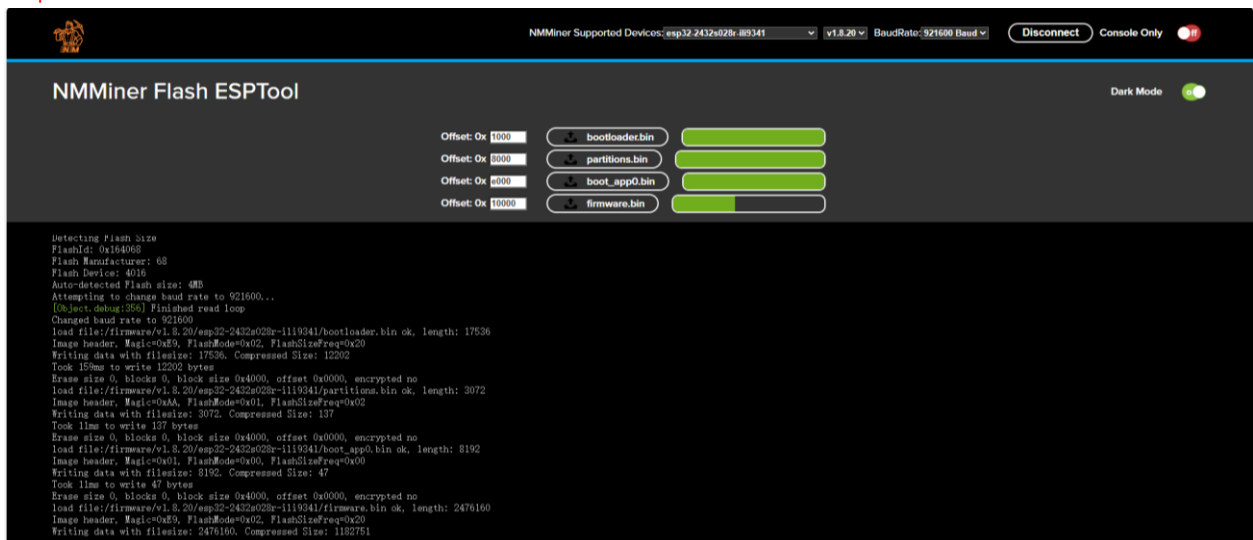
4. Click **“Connect and Program”**, select the correct port in the pop-up window, and click **“Connect”**.

Please note:

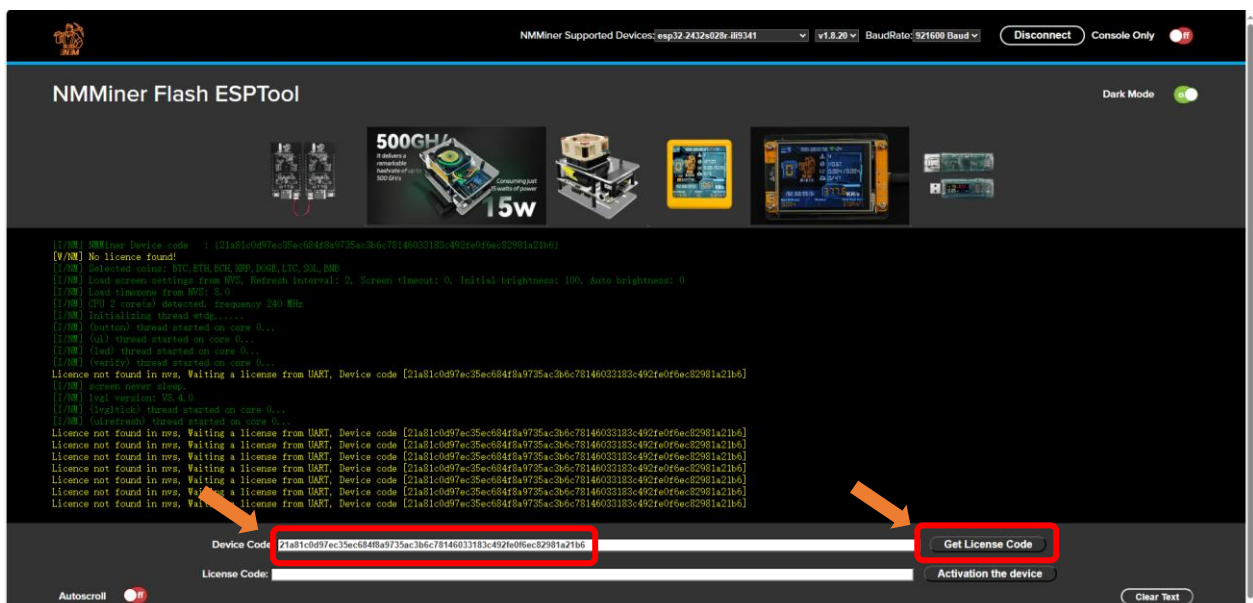
- The port may vary among computers
- Generally, COM1 is not the serial port of the ESP32.



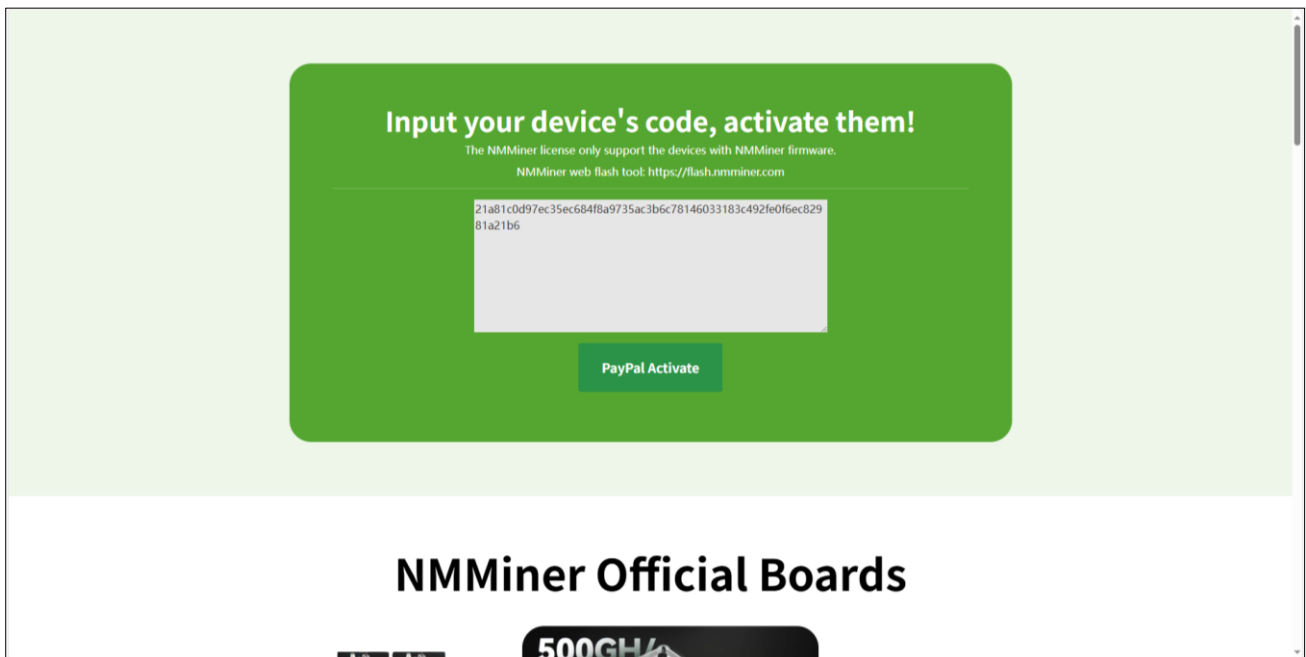
5. The firmware starts to flash into the Freenove ESP32 Display. **Do NOT remove power supply during this process.**



6. Click **“Get License Code”** after the firmware successfully flashes.

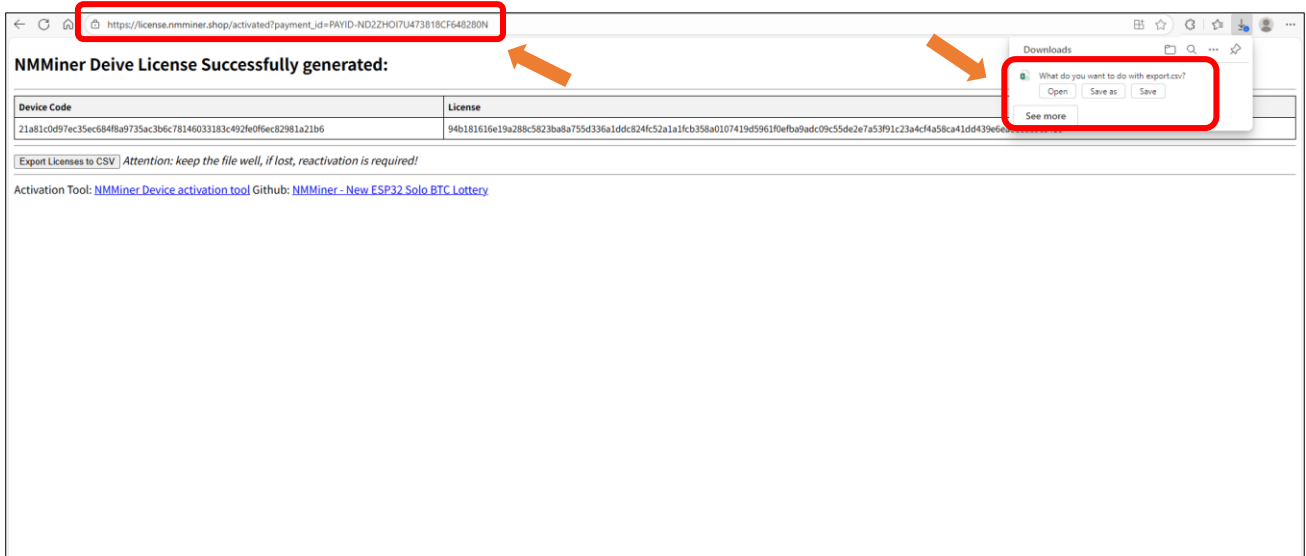


7. Click “PayPal Activate” to make payment.



8. Please securely store the generated License (important). **If the License is lost, it will need to be repurchased.**

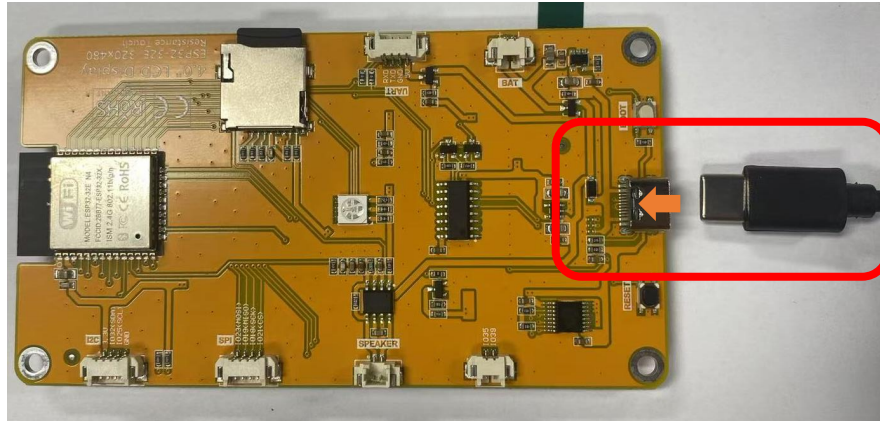
When generating the license, the website will automatically download a CSV file containing the License information. You may also save the current webpage URL, as it can be reopened later.



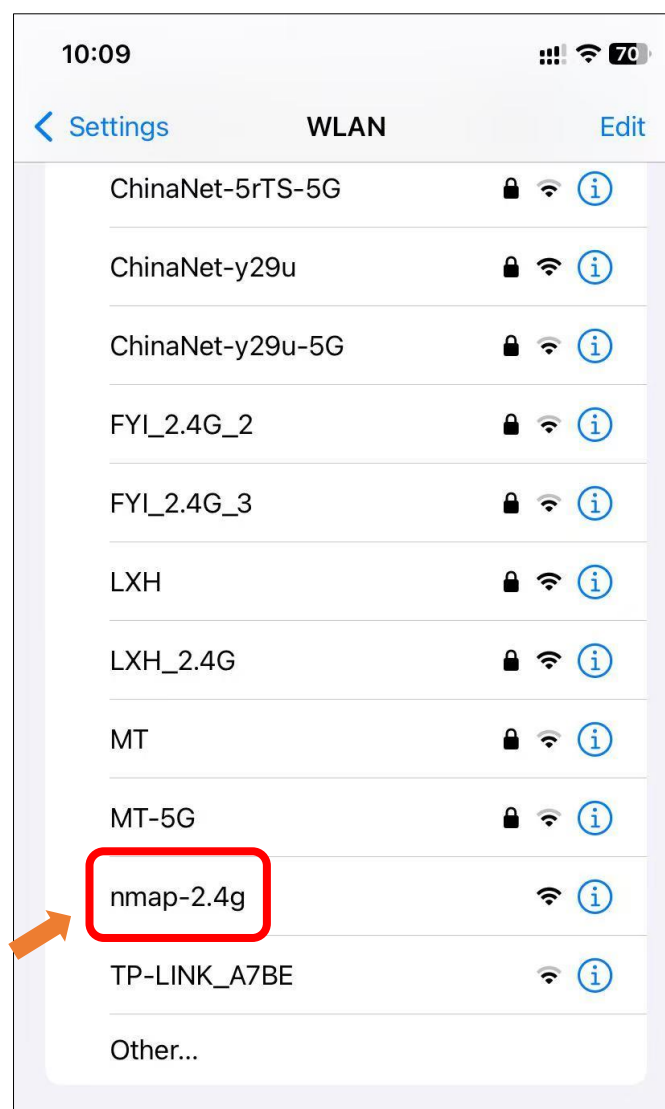
Please note: Each device requires a unique license code for activation. Therefore, you will need to purchase a separate activation code for every development board you plan to use in order to complete the activation process.

Wi-Fi Configuration

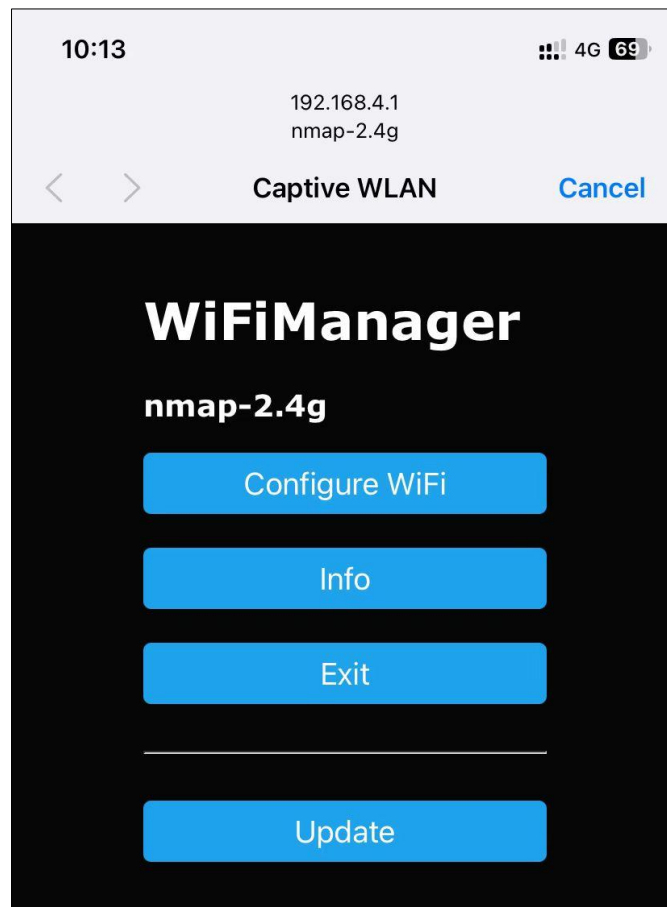
1. Connect the Freenove ESP32 Display to your computer. If the firmware has not been flashed yet, please navigate back to the [previous section](#) to do it.



2. Turn ON WLAN, locate and click “nmap-2.4g”.



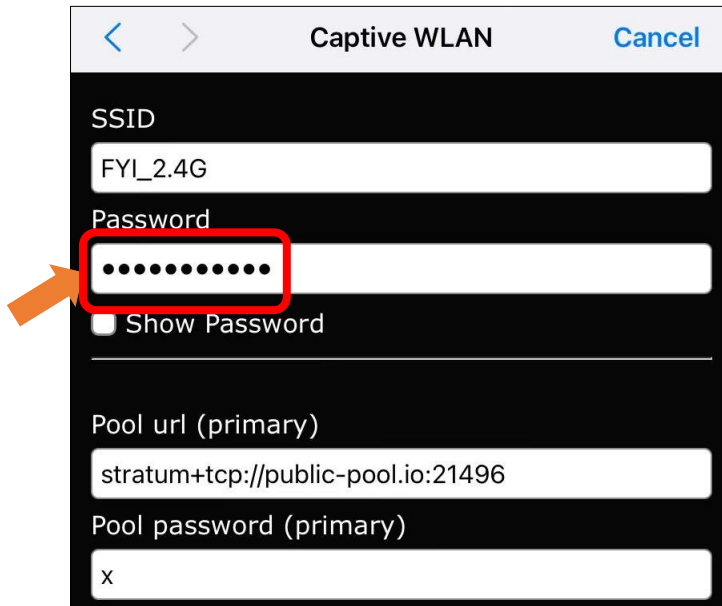
3. Wait for it to pop up the following interface, click "Configure WiFi".



4. Select your network. Please note that ESP32 **only supports 2.4GHz network**.



5. Enter the correct WiFi password.



< > Captive WLAN Cancel

SSID
FYI_2.4G

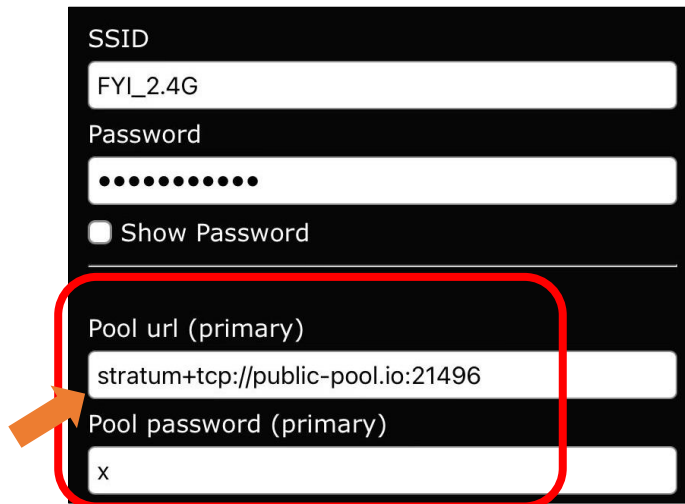
Password
.....

☐ Show Password

Pool url (primary)
stratum+tcp://public-pool.io:21496

Pool password (primary)
x

6. Enter the mining pool information. If the mining efficiency is unsatisfactory, you can switch to [other mining pools](#) as needed.



SSID
FYI_2.4G

Password
.....

☐ Show Password

Pool url (primary)
stratum+tcp://public-pool.io:21496

Pool password (primary)
x

7. Enter the correct BTC address. If you do not have a BTC address yet, please refer to [Obtaining the BTC Receiving Address](#) Section to get it.



Pool url (primary)
stratum+tcp://public-pool.io:21496

Pool password (primary)
x

BTC address (primary)
bc1qa5hcrn0g25gt6t30qp66y29k6h5fw2ztu4l0gu

Pool url (fallback)
stratum+tcp://pool.tazmining.ch:33333

8. Enter the correct time zone.



Pool password (fallback)

x

BTC address (fallback)

bc1qa5hcrn0g25gt6t30qp66y29k6h5fw2ztu4l0gu

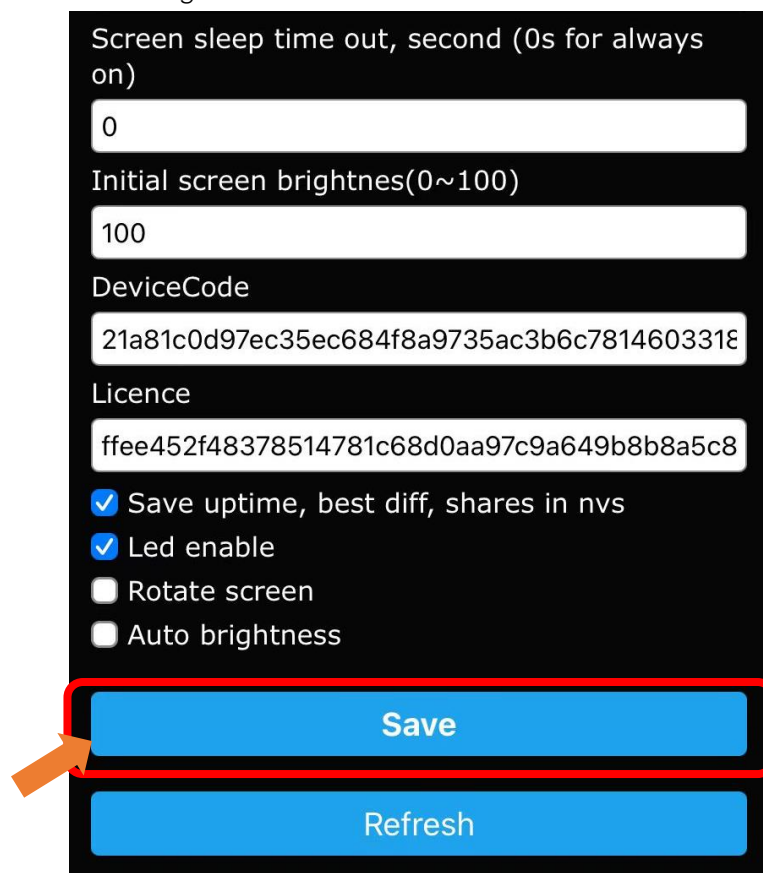
Time zone (UTC -12/+12)

8.00

UI refresh interval(second)

2

9. Click "Save" to save the configuration.



Screen sleep time out, second (0s for always on)

0

Initial screen brightness(0~100)

100

DeviceCode

21a81c0d97ec35ec684f8a9735ac3b6c7814603318

Licence

ffee452f48378514781c68d0aa97c9a649b8b8a5c8

☒ Save uptime, best diff, shares in nvs

☒ Led enable

☐ Rotate screen

☐ Auto brightness

Save

Refresh

10. Now the NMMiner project begins to work.

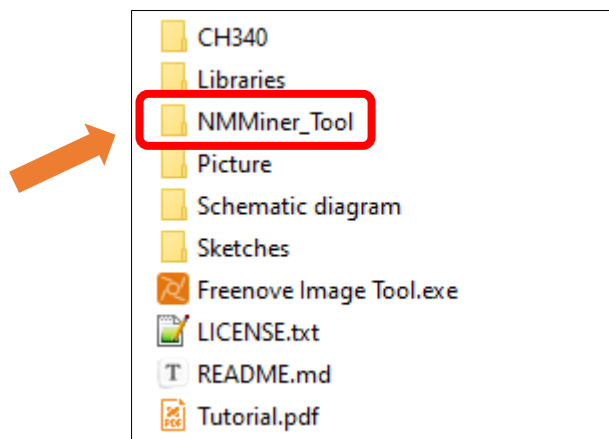


Management Tools

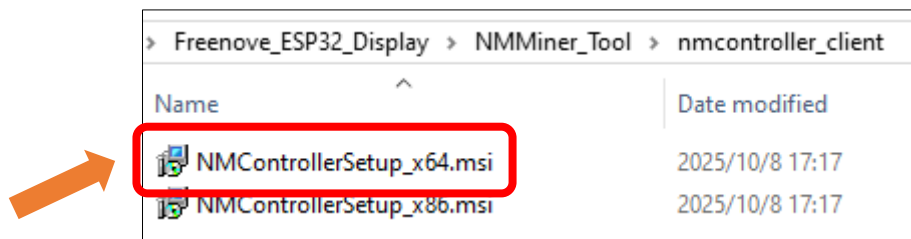
NMController Client

Note: NMController Client only works on Windows.

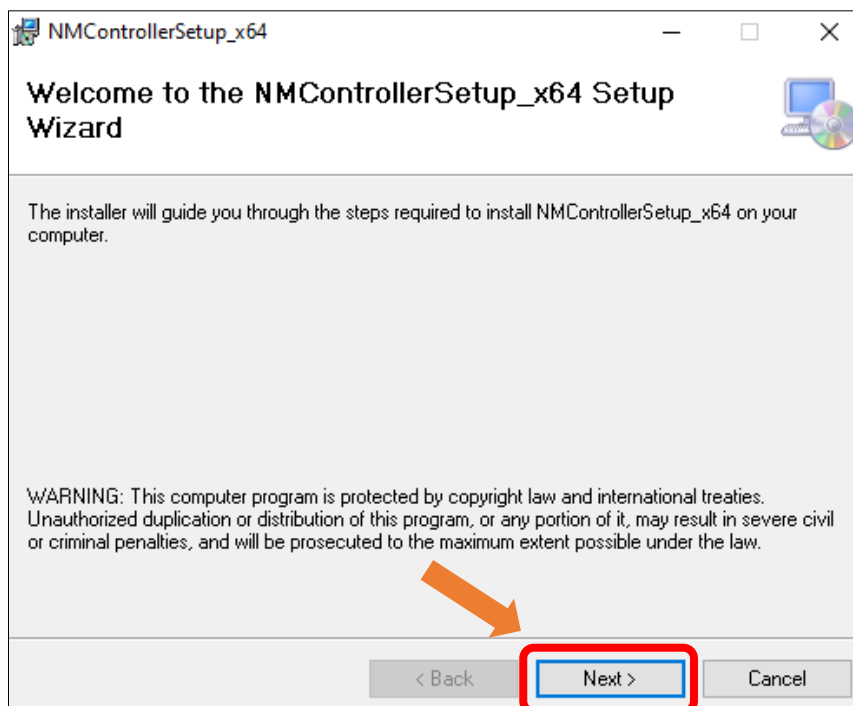
1. Open “NMMiner_Tool/nmcontroller_client”.



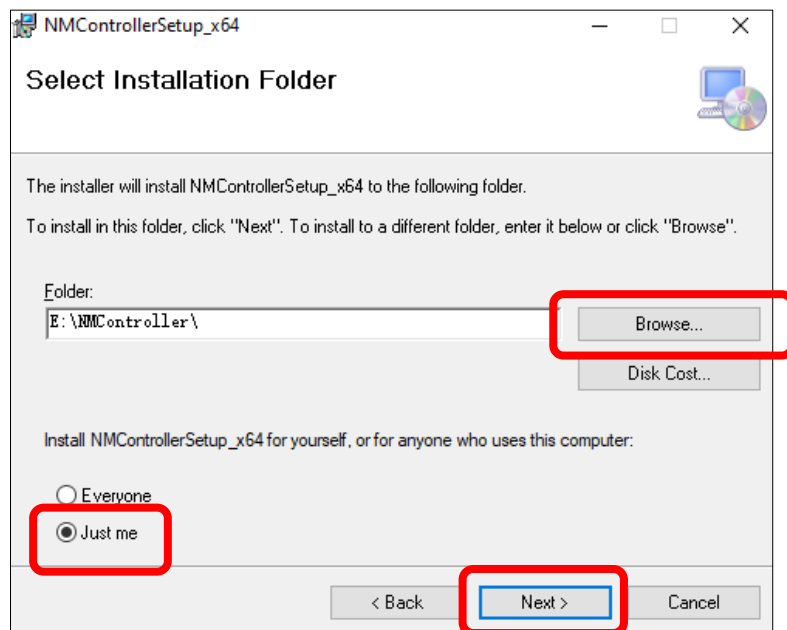
2. Double click “NMControllerSetup_x64.msi” to open it.



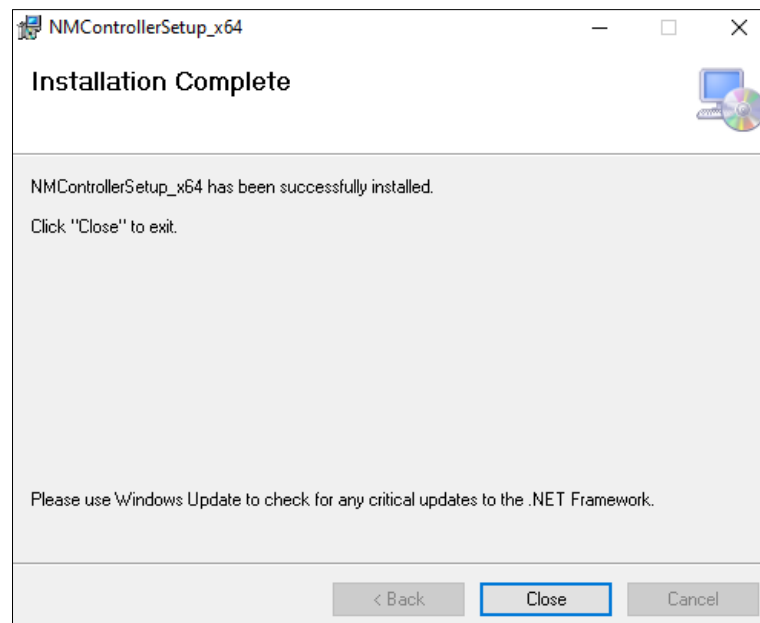
3. Click “Next”.



4. Click **"Browse"** to select the installation location, check **"Just me"**, and click **"Next"**.



5. Close the window when the installation completes.



Open the installed NMController Client and you can see the current device data.

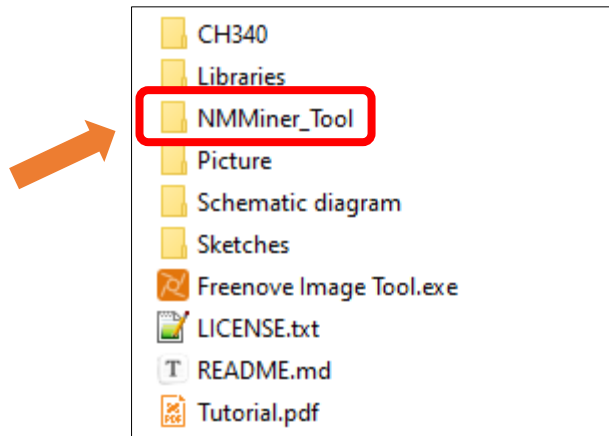
NMController v0.4.1												
IP	BoardType	HashRate	Share	Pool	NetDiff	LastDiff	BestDiff	Valid	Temp	RSSI(dBm)	FreeHeap	Version
192.168.1.26	cyd 2.8	999.55KH/s	0/162/100.0%	public-pool.io:21496	146.7T	0.010	23.99 23.99	0	0.0	-37	98.20	v1.8.20

If you have any concerns, please feel free to contact us via support@freenove.com

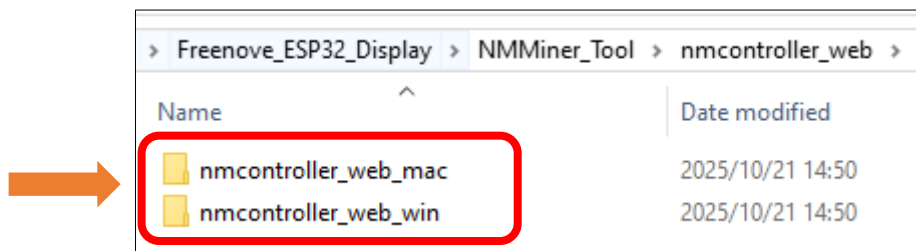
NMController Web

NMController Web: A cross-platform web client designed with Python, compatible with both Windows and macOS systems.

1. Open "NMMiner_Tool/nmcontroller_web".



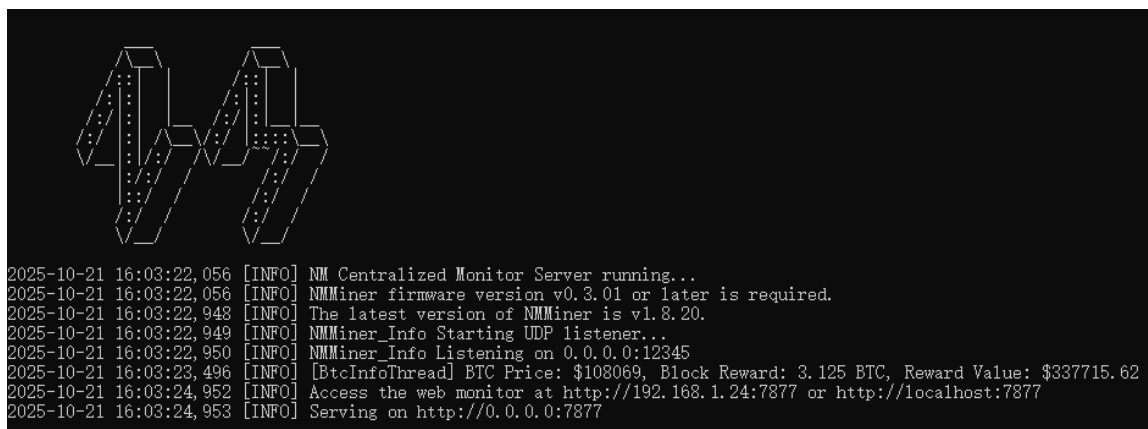
2. Select the folder based on your computer's OS.



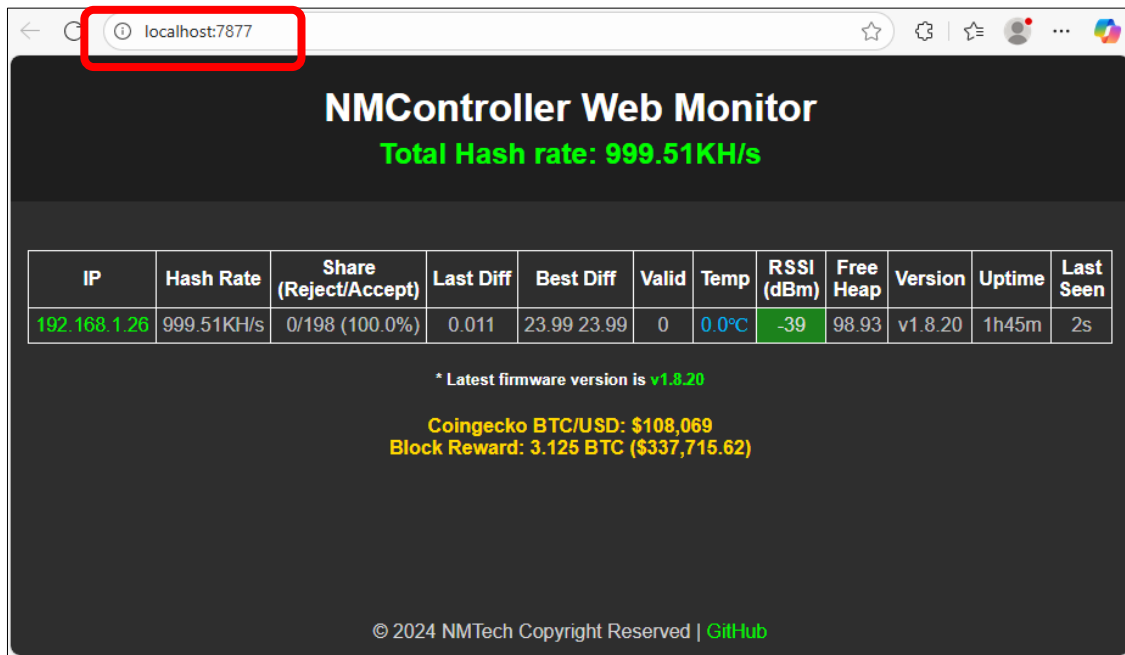
3. Double click to run the program.
 - (1) You can access the management interface by visiting <http://localhost:7877> on the current device.
 - (2) Devices in the same LAN can access the management interface by visiting <http://192.168.1.24:7877>

Notes:

- The following interface must NOT be closed when using NMController Web.
- The COM port may vary among devices. Please refer to the printed information for the correct port number.

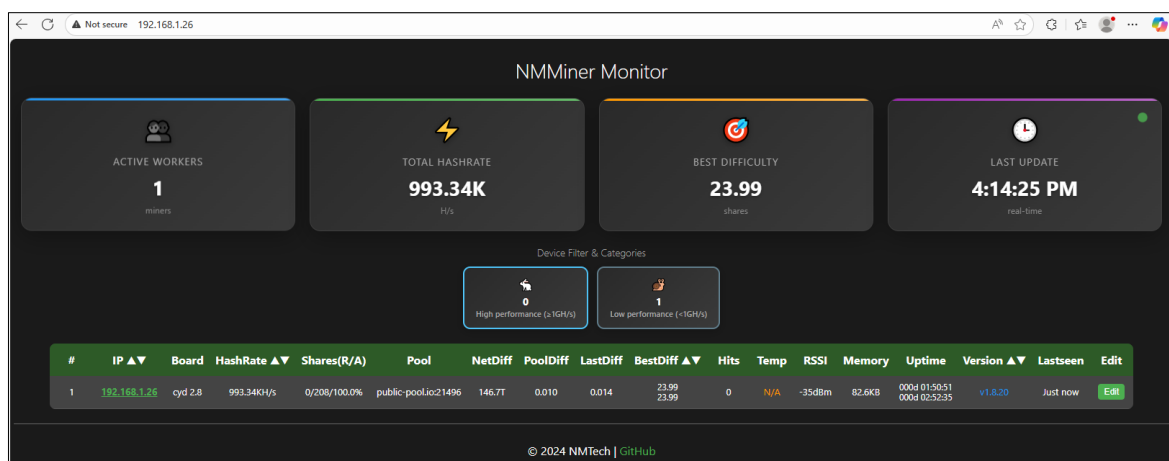


4. Access the corresponding address to open the management tool.



IP Address Management

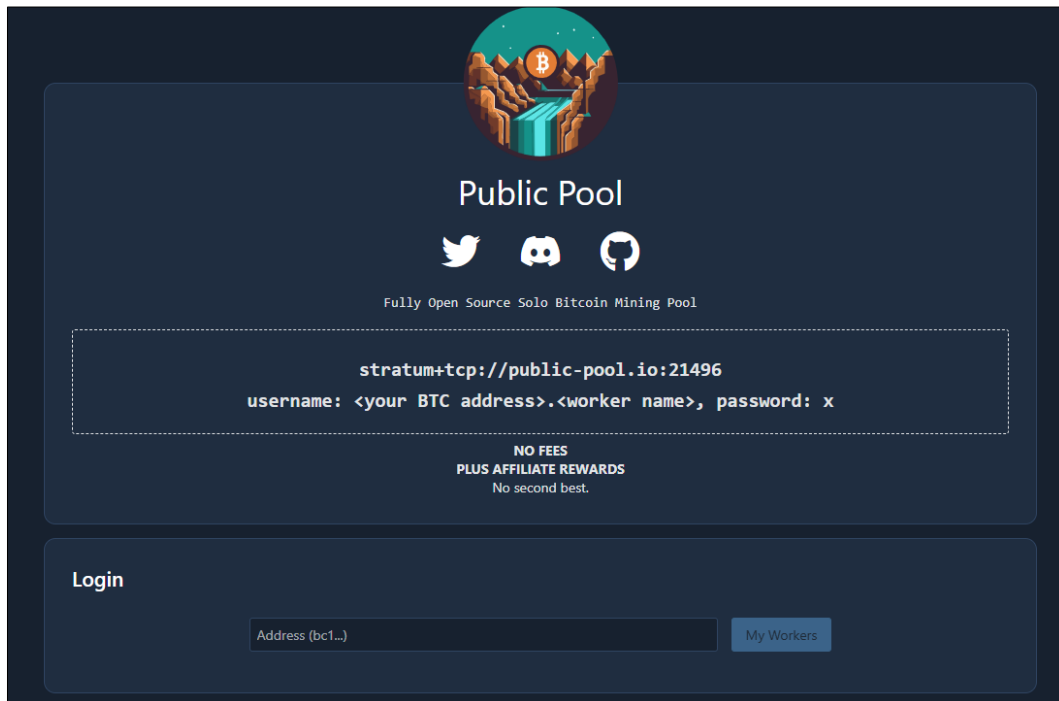
In the NMMiner interface, there is an IP address displayed. Devices connected to the same local network can access this IP address to open the management interface.



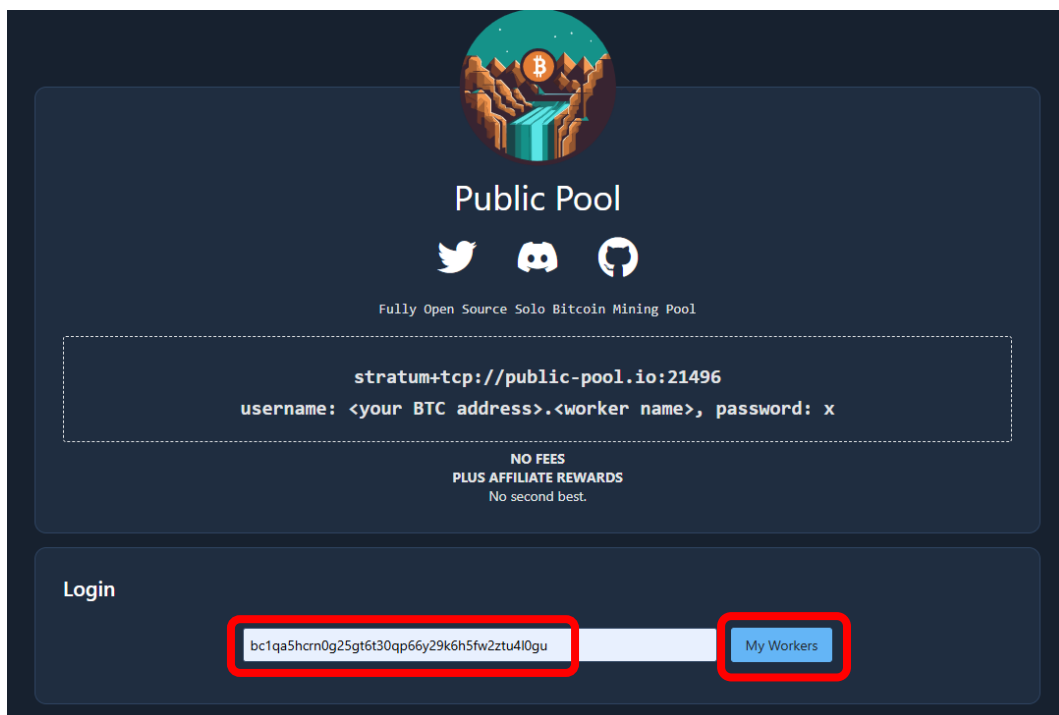
Public Pool

When the mining pool public-pool.io:21496 is selected, you can manage it via the web interface using the link (<https://web.public-pool.io/#/>).

1. Open public-pool



2. Enter BTC receiving address, and click "My Workers".



3. You can view the current working status of the worker.



Please note: This webpage only detects and manages miners currently working in the Public Pool that use your BTC address as the login or payout address. Once the pool is changed, this webpage will no longer be accessible for management.

NerdMiner_v2

This project utilizes Freenove ESP32 Display and NerdMiner_v2 to implement Bitcoin mining. It requires some programming background and a basic understanding of digital currencies.

Project Introduction

NerdMiner_v2 (https://github.com/BitMaker-hub/NerdMiner_v2) is an excellent open-source BTC mining firmware designed for ESP32 development boards. Its goal is to allow users to experience the process of "mining a Bitcoin block" using a small piece of hardware. The primary aim of this project is to help more people **learn about cryptocurrency mining** while owning an aesthetically pleasing desktop accessory.

NerdMiner_v2 supports mining on **solo pools** by implementing the **Stratum protocol** (by default, it connects to **public-pool.io**, which supports Nerdminer devices, but the pool address can be customized). Freenove has ported this project to run on the Freenove ESP32 Display.

This tutorial will guide you through the process of running the project on the Freenove ESP32 Display.

Precautions

- **Cost**
 - As of the time of writing this document, NerdMiner_v2 is temporarily free to use.
- **Security Precautions**
 - Never disclose your wallet's private key or recovery phrase to anyone. The only information you need to provide is your public cryptocurrency receiving address.
- **Troubleshooting**
 - If you have followed the tutorial but are still unable to use the miner, please feel free to contact us for assistance. (support@freenove.com)

Compatibility Description

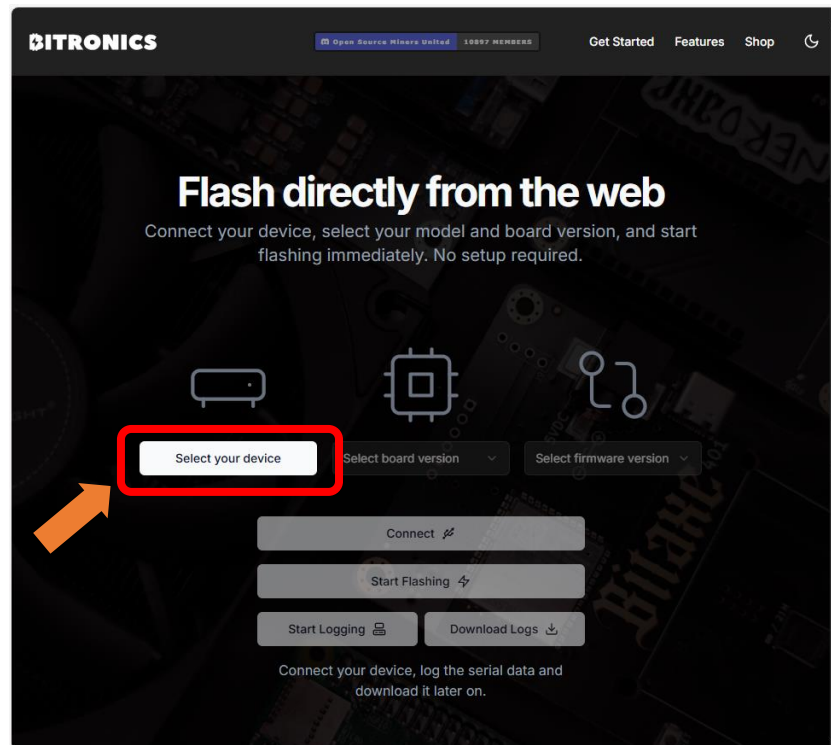
At the time of writing this guide, the Freenove ESP32 Display is available in five different models. Although they vary in terms of screen drivers, resolutions, and screen sizes, all of them can be set up using this tutorial to learn how to use NerdMiner_v2.

Models	Specifications		Effect
FNK0103F_2P8	Size	2.8 inch	Color Inversion by Default
	Resolution	240x320	
	Driver	ILI9341	
FNK0103B_2P8	Size	2.8inch	Fully Compatible
	Resolution	240x320	
	Driver	ST7789	
FNK0103L_3P2	Size	3.2inch	Incorrect Color
	Resolution	240x320	
	Driver	ST7789	
FNK0103N_3P5	Size	3.5inch	Not Full Screen
	Resolution	320x480	
	Driver	ST7796	
FNK0103S_4P0	Size	4.0inch	Not Full Screen
	Resolution	320x480	
	Driver	ST7796	

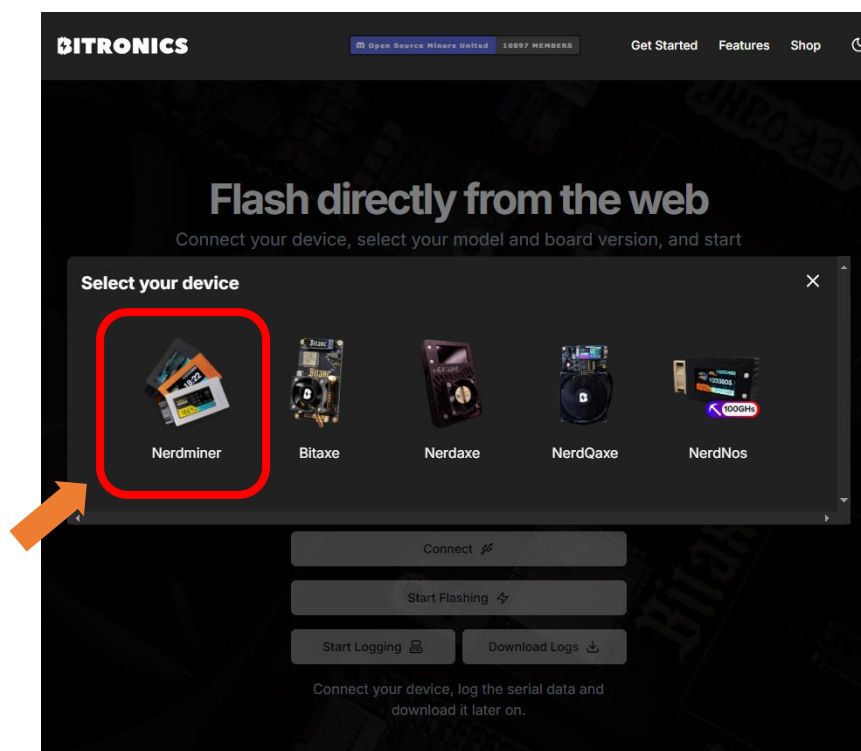
How to Use

Firmware Flashing

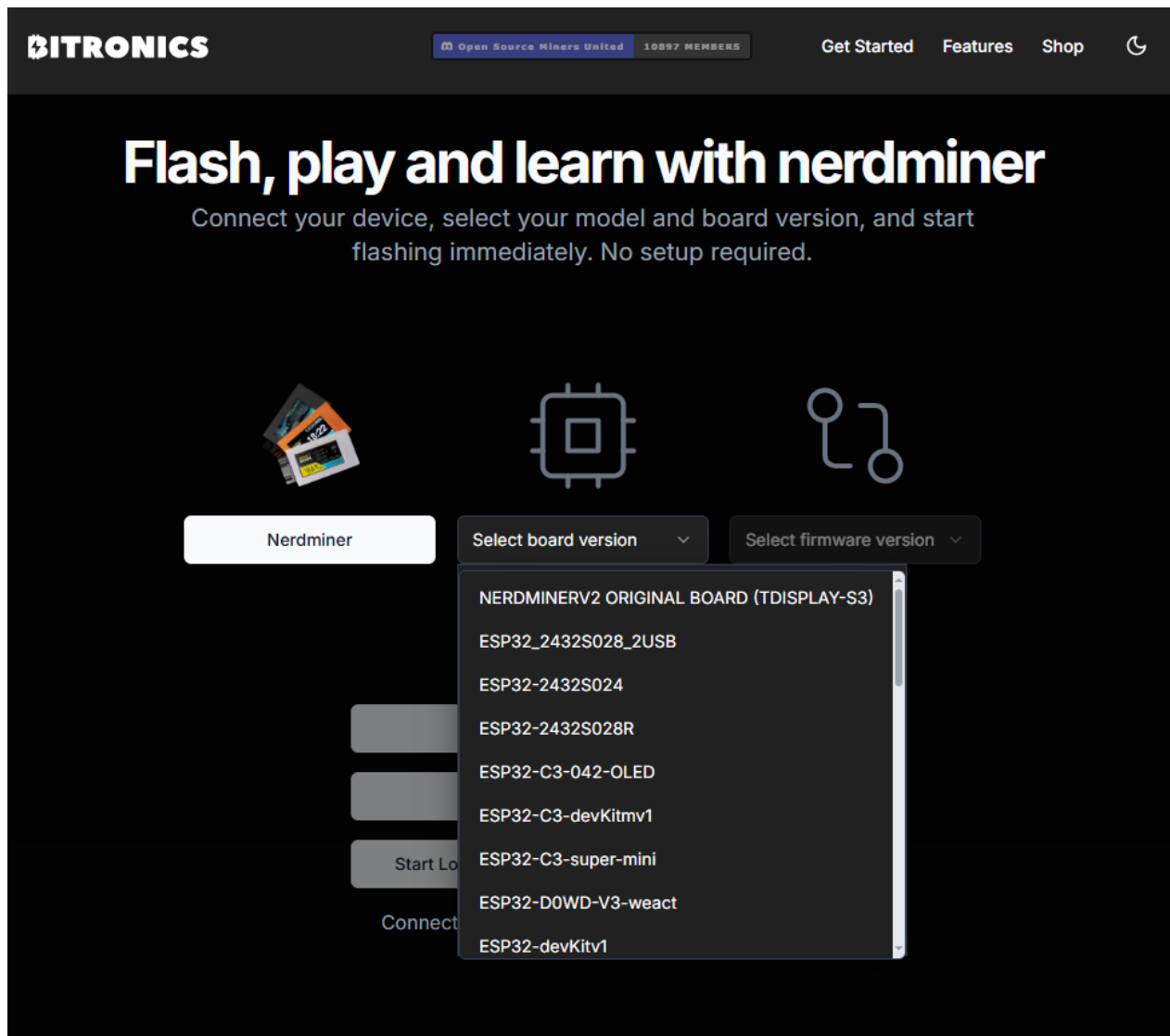
1. Open the [Bitronics flasher](#) interface and click “Select your device”.



2. Select “Nerdminer”.



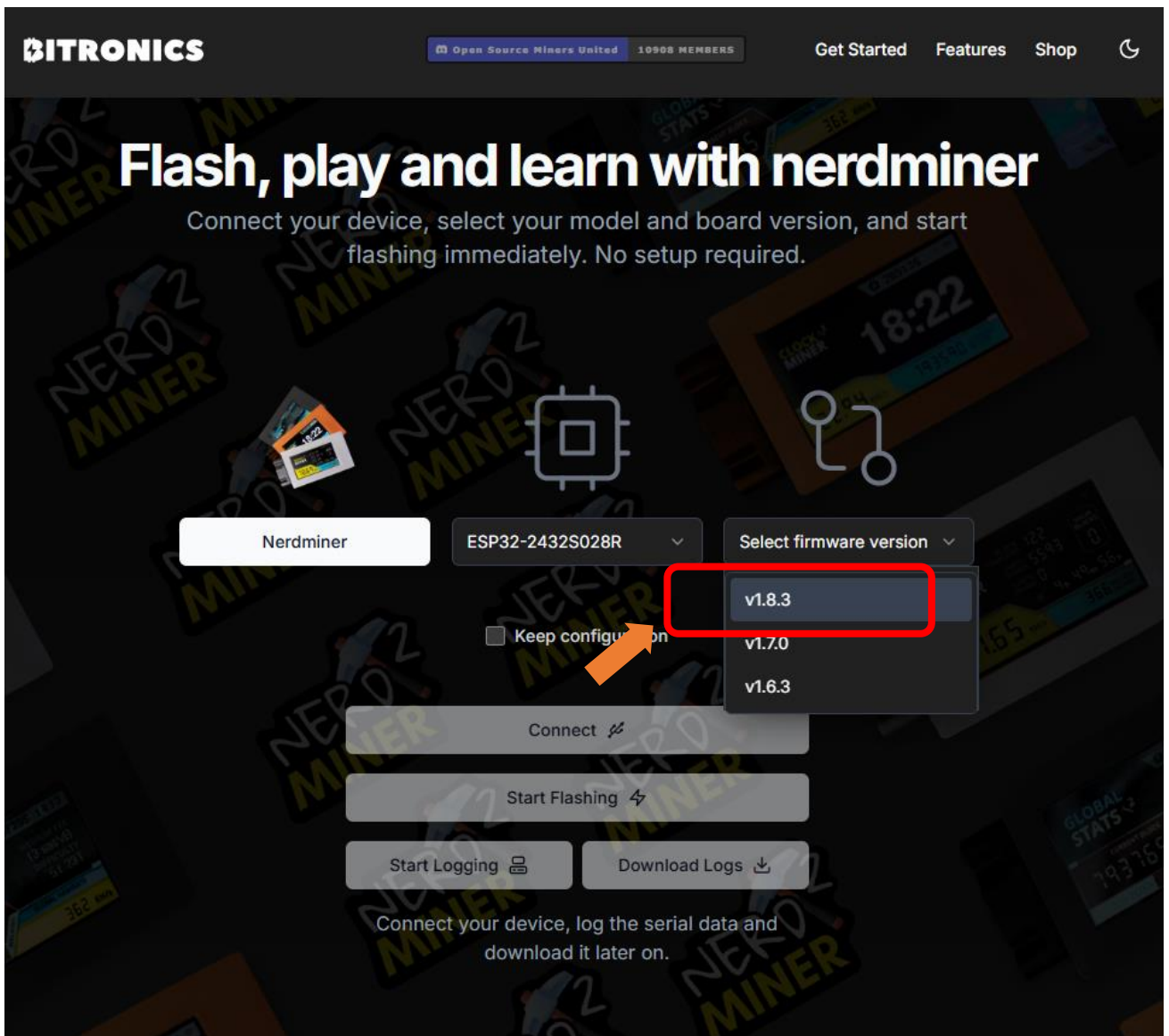
3. Select the correct firmware for your display based on the table below.



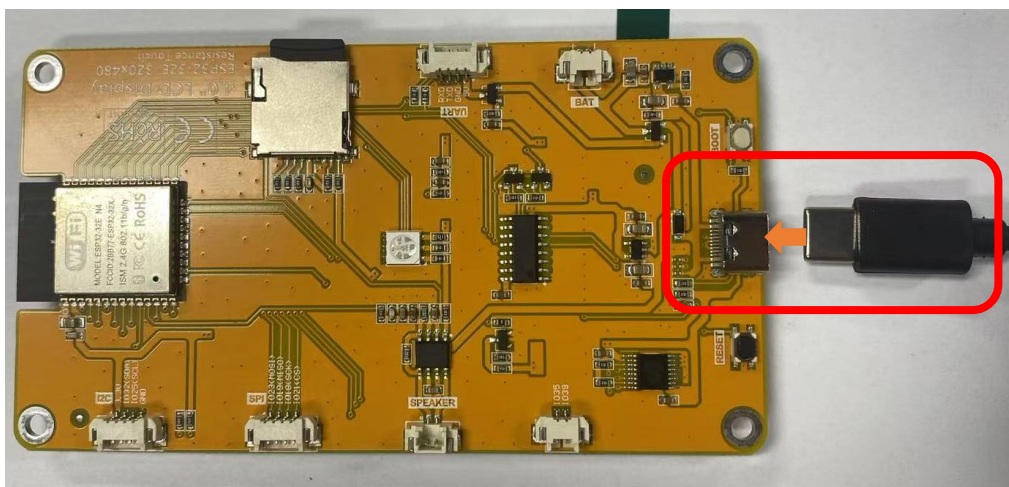
Models	Board Version
FNK0103F_2P8	ESP32-2432S028R
FNK0103B_2P8	ESP32-2432S028R
FNK0103L_3P2	ESP32-2432S024
FNK0103N_3P5	ESP32-2432S024
FNK0103S_4P0	ESP32-2432S024

If you have any concerns, please feel free to contact us via support@freenove.com

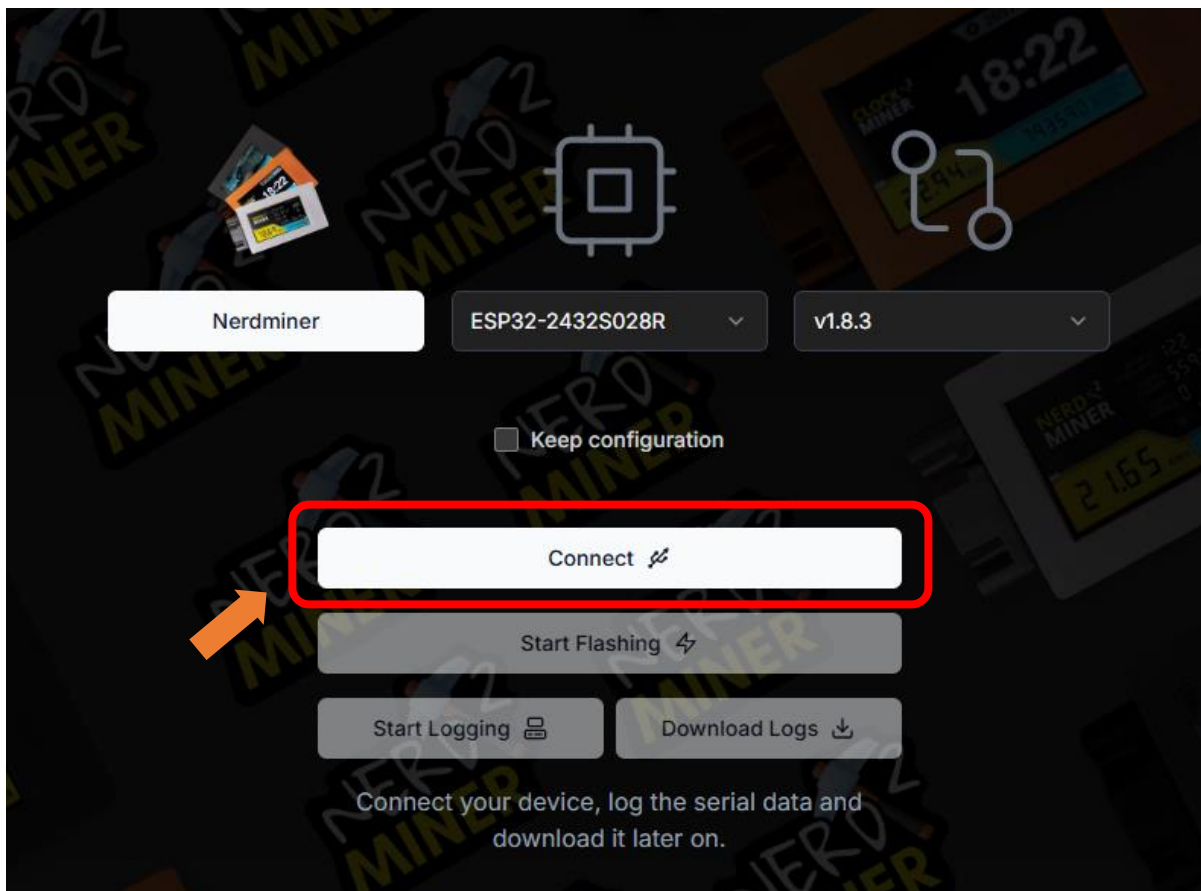
4. Select the latest firmware version.



5. Connect the Freenove ESP32 Display to your computer.



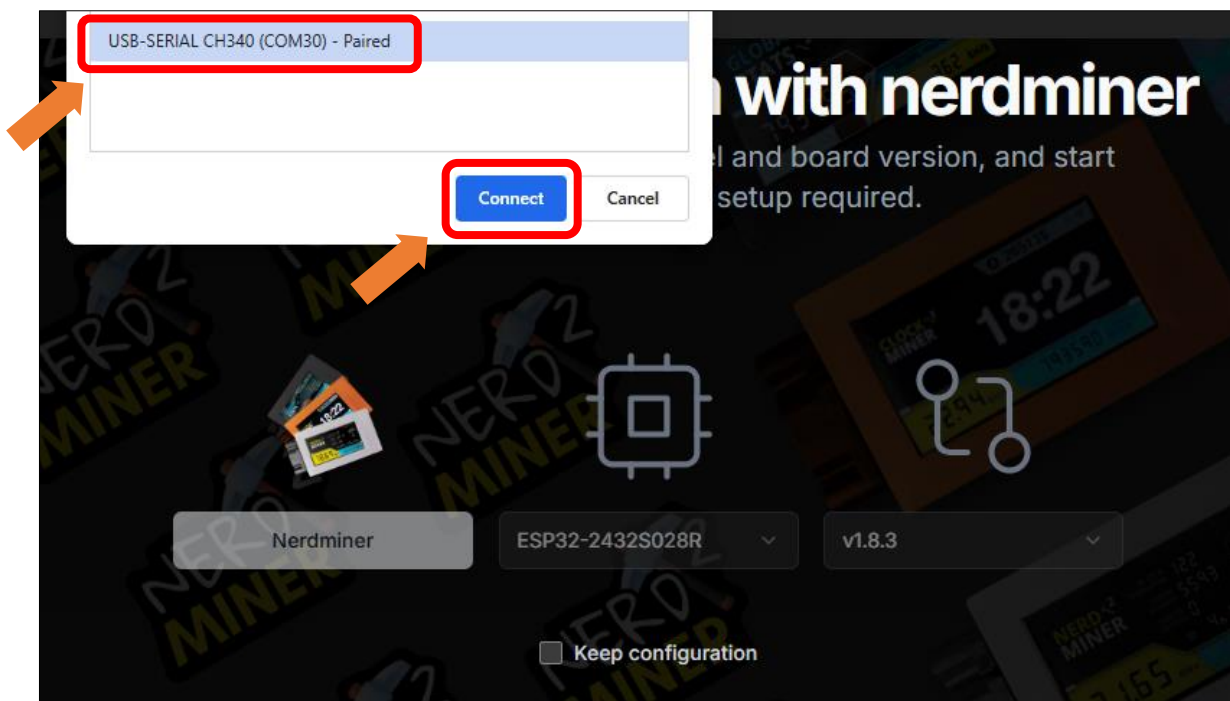
6. Click **“Connect”**.



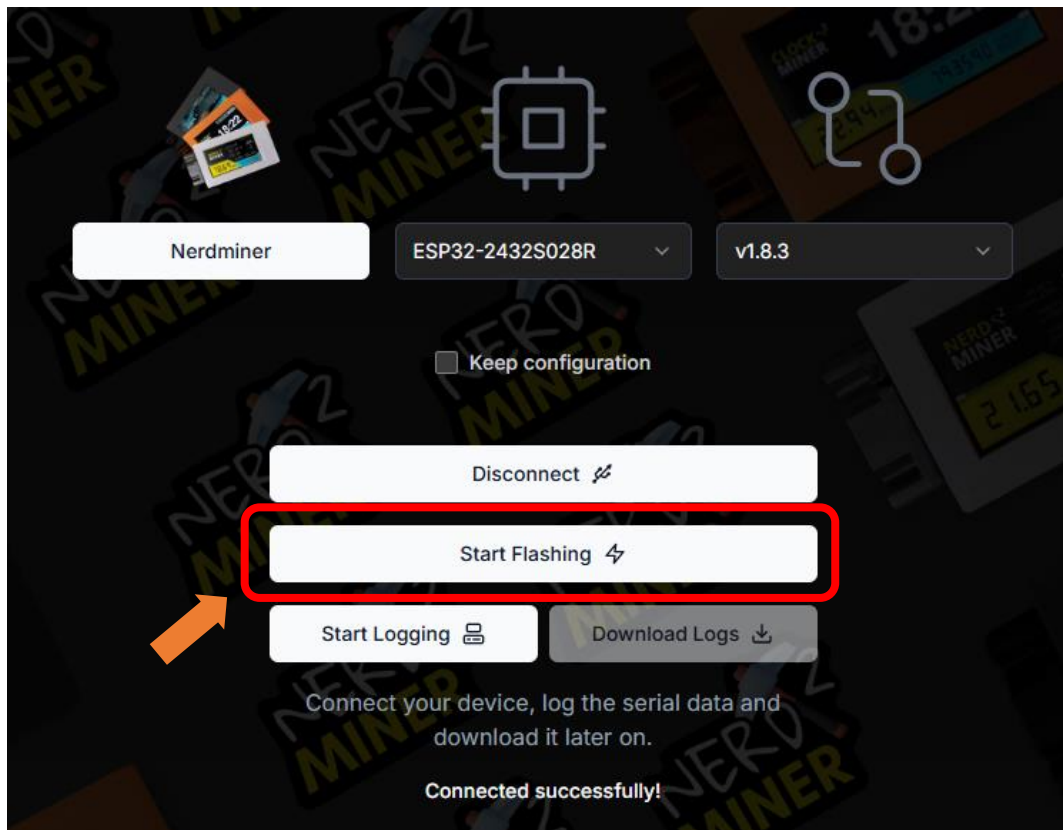
7. Select the port of your ESP32.

Notes:

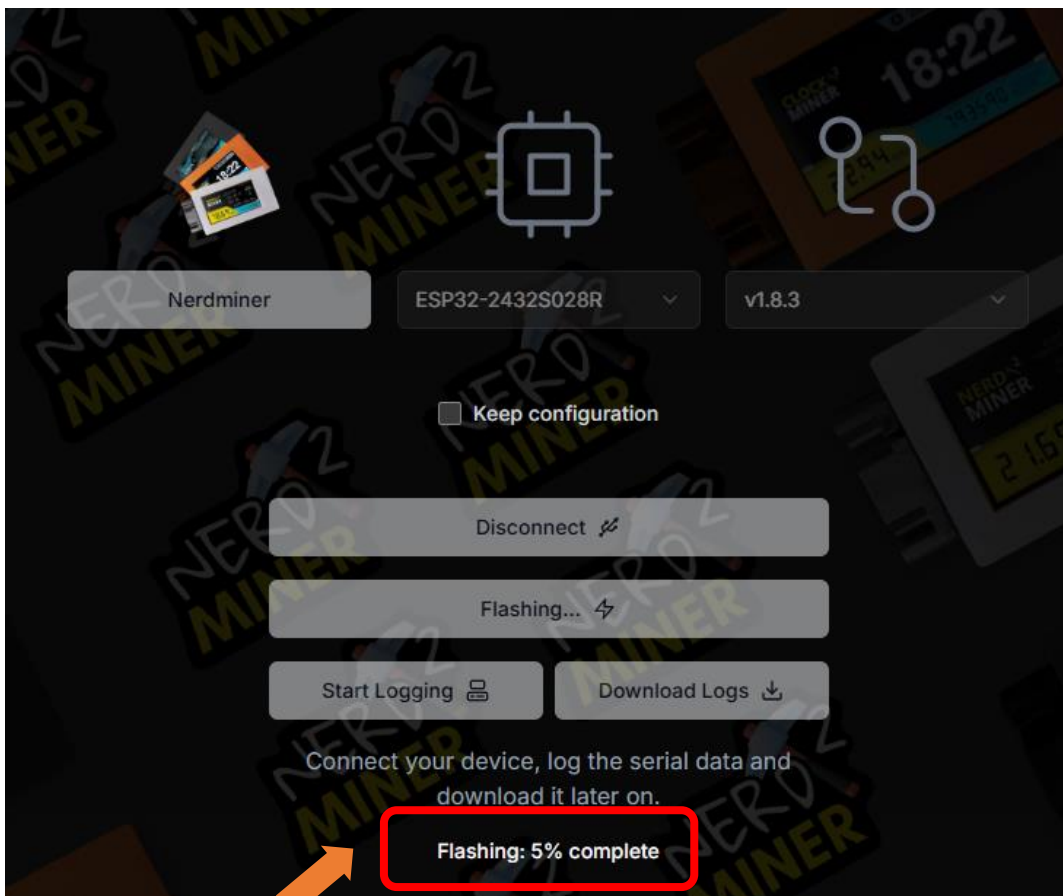
- The port number may vary among devices.
- Generally, COM1 is not the port of ESP32.



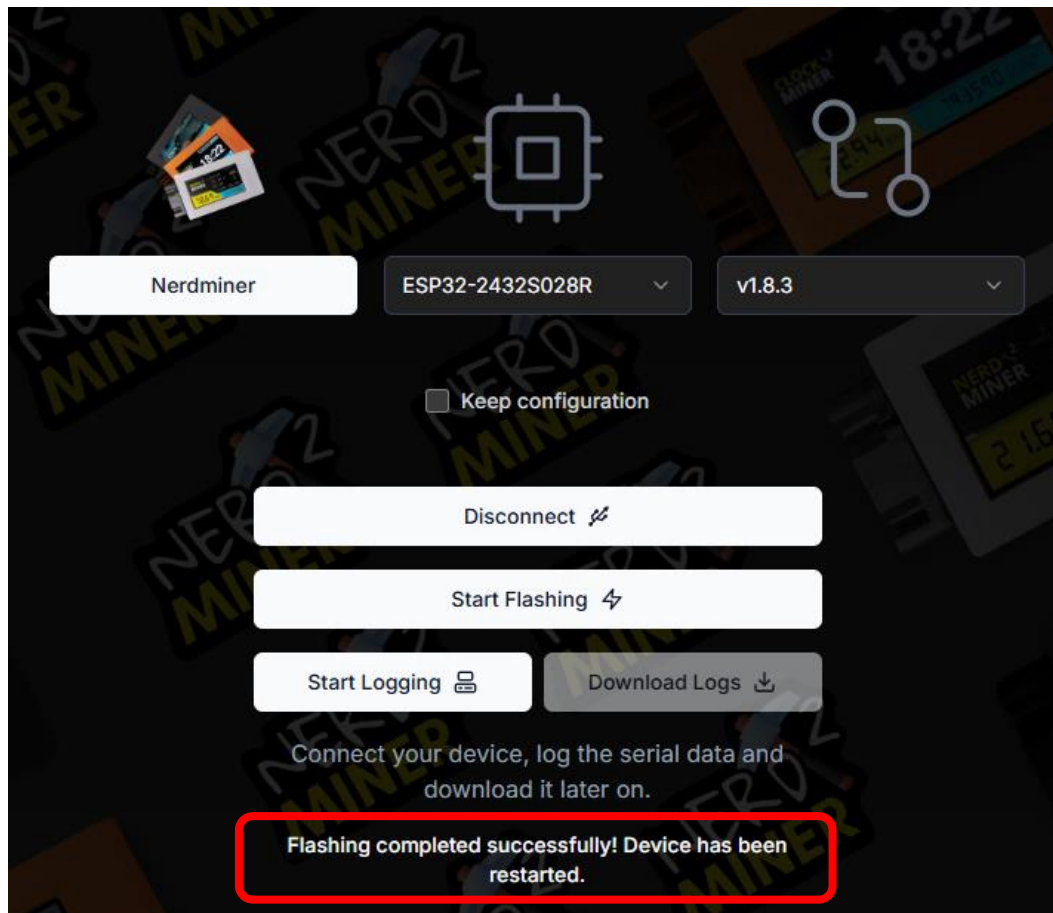
8. Click "Start Flashing".



9. The firmware starts to flash into the Freenove ESP32 Display. **Do NOT remove power supply during this process.**

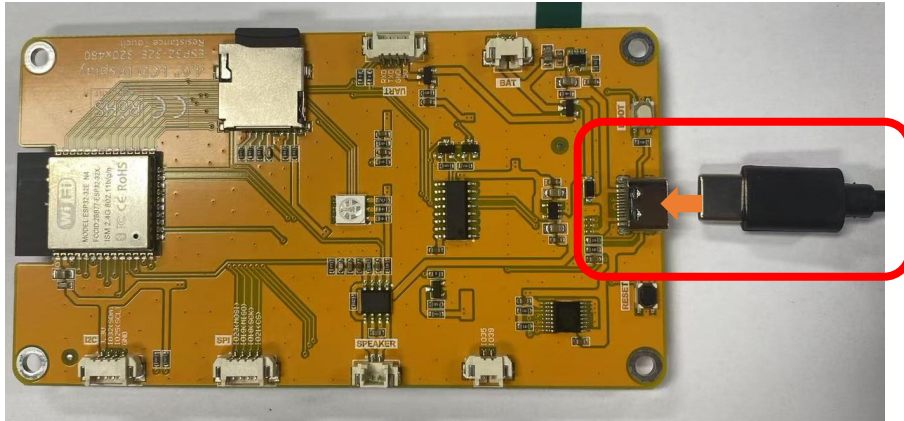


10. The following message indicates successfully firmware flashing.



Wi-Fi Configuration

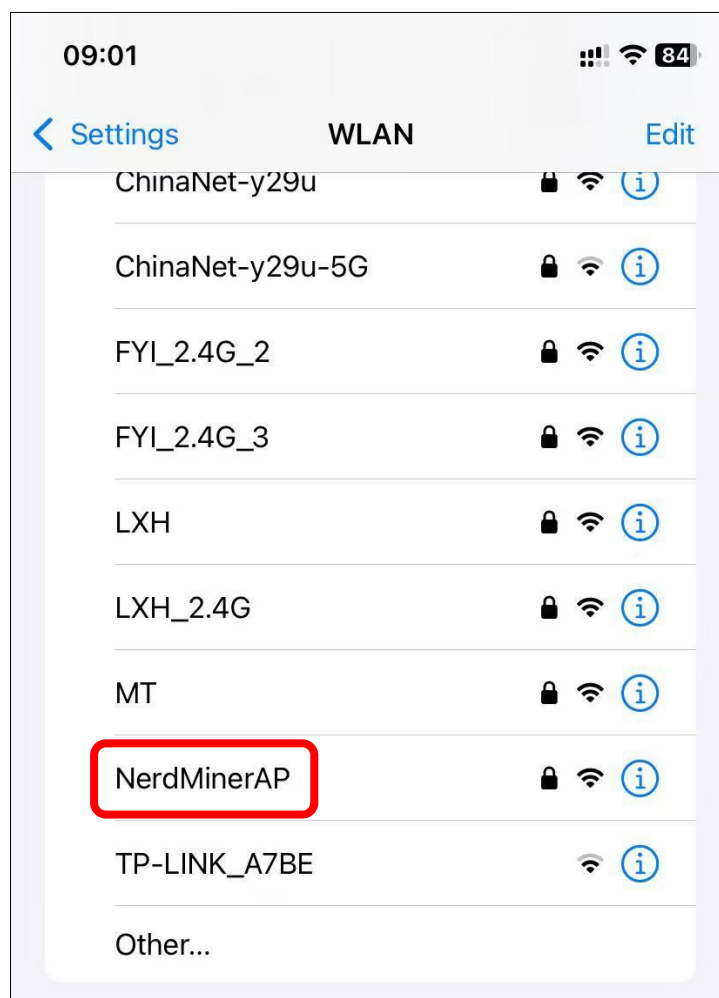
1. Connect the Freenove ESP32 Display to your computer. If the firmware has not been flashed yet, please navigate back to the [Firmware Flashing Section](#) to do it.



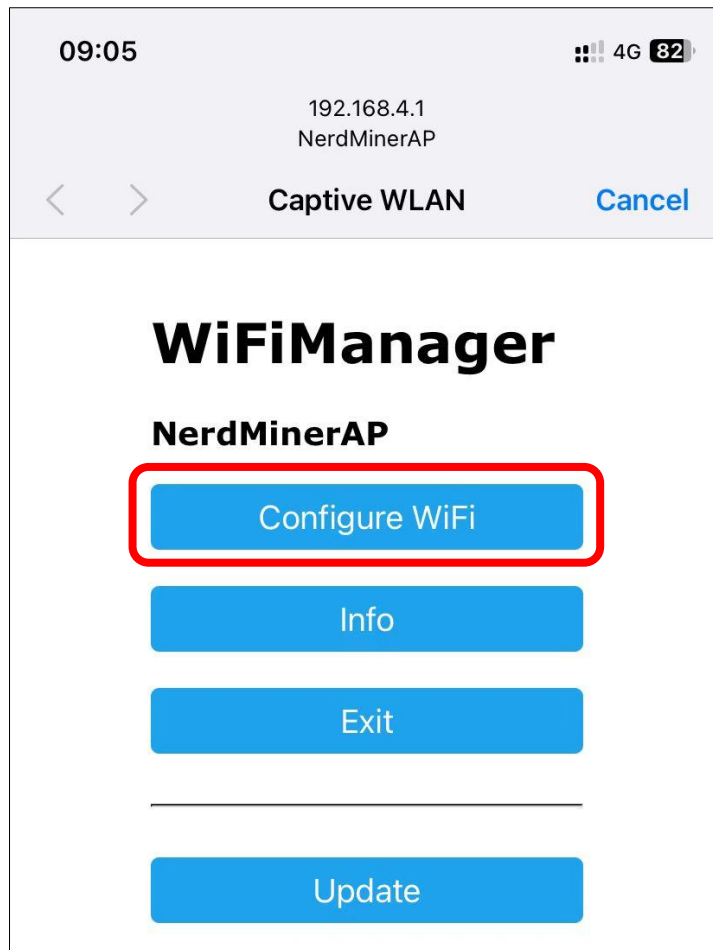
2. Turn ON WLAN and connect to **NerdMinerAP**

WiFi SSID: **NerdMinerAP**

PASSWORD: **MineYourCoins**



3. Click "Configure WiFi"



4. Select your network. Please note that ESP32 **only supports 2.4GHz network**.



5. Configure device information.
6. If you do not have a BTC address yet, please refer to [Obtaining the BTC Receiving Address](#) Section to get it.
7. If the mining efficiency is unsatisfactory, you can switch to [other mining pools](#) as needed.

09:08 192.168.4.1 NerdMinerAP

< > Captive WLAN Cancel

SSID
FYI_2.4G

Input the correct WiFi password

Password
●●●●●●●●

☐ Show Password

Change mining pool if needed

Pool url
public-pool.io

Pool port
21496

Input your BTC address

Your BTC address
bc1qa5hcrn0g25gt6t30qp66y29k6h5fw2ztu4l0gu

Configure the correct timezone

TimeZone fromUTC (-12/+12)
2

Features

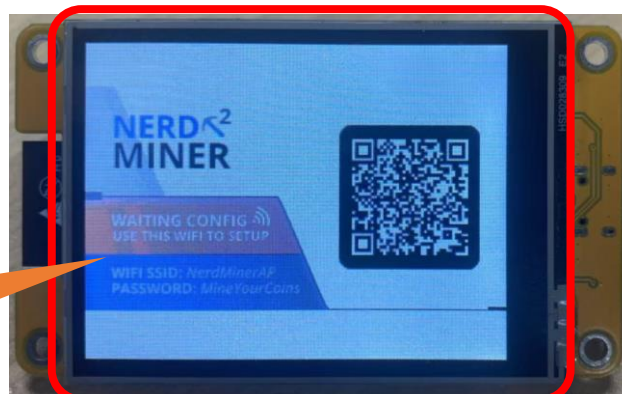
☐ Save mining statistics to flash memory.

☒ Invert Display Colors (if the colors looks weird)Screen backlight Duty Cycle (0-255)

250

If color inversion occurs as shown below, check here.

Color inversion

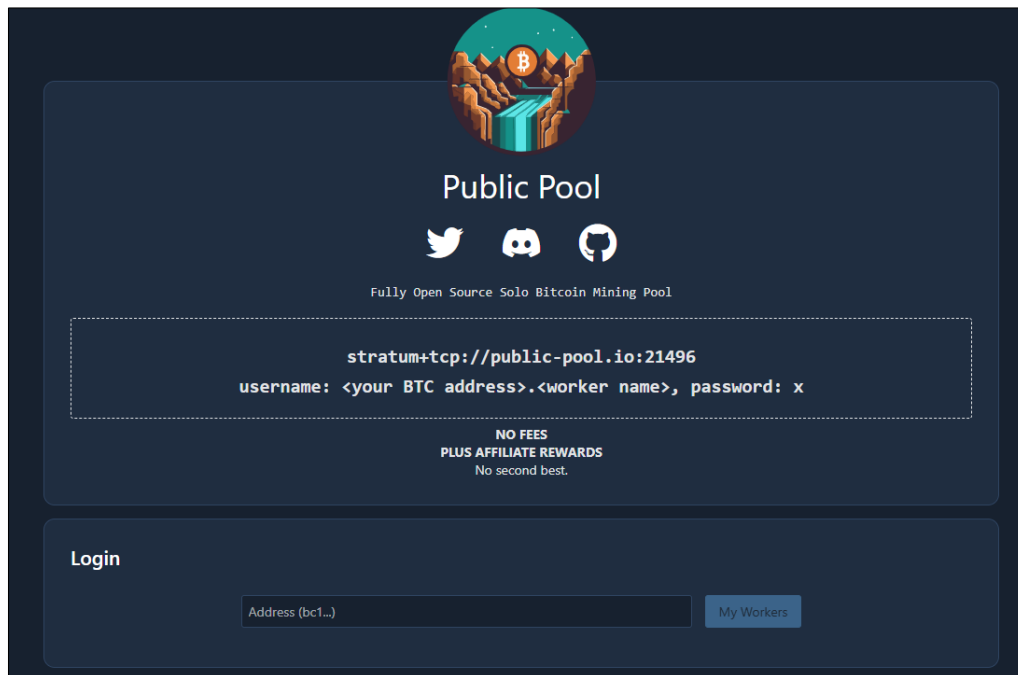


Management Tool

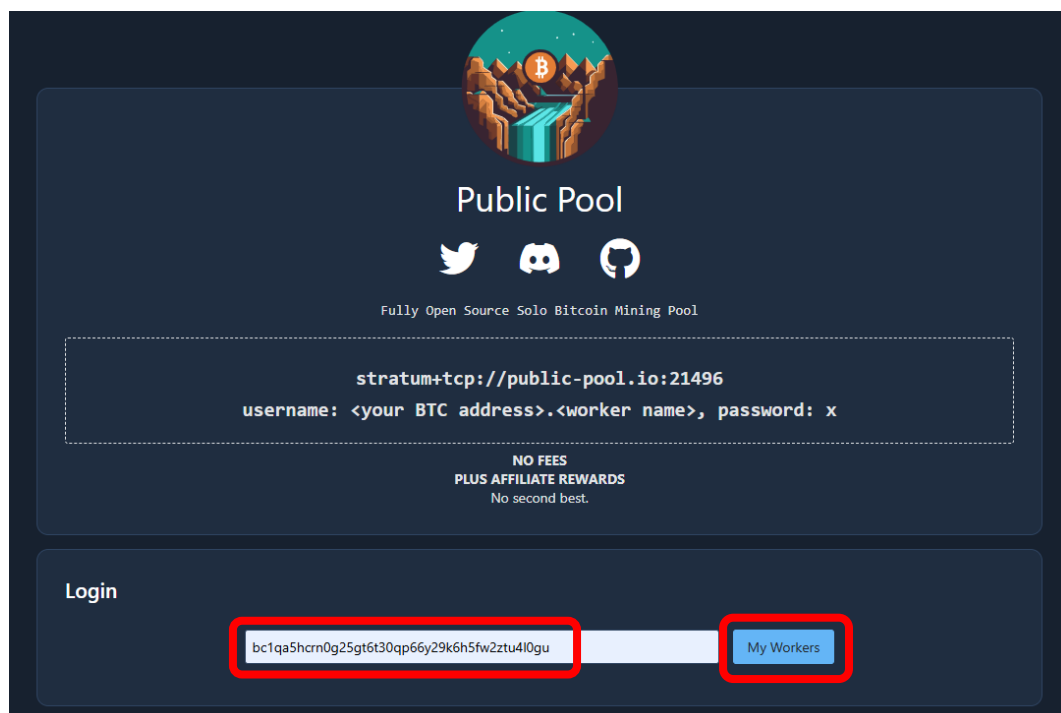
Public Pool

When using the pool public-pool.io:21496, you can manage it via the web link <https://web.public-pool.io/#/>.

1. Open public-pool



2. Enter your BTC receiving address, and click "My Workers".



3. You can view the current working status of the worker.



Please note: This webpage only detects and manages miners currently working in the Public Pool that use your BTC address as the login or payout address. Once the pool is changed, this webpage will no longer be accessible for management.